Records too sharp to discipline: A governance externality in competence disclosure

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Abstract

Records of expert performance mix competence with conduct. In markets such as health care and financial advice, one institution often decides how finely the record is kept while another decides how much competence to disclose. We study how this split shapes conduct information. Sharper disclosure deconfounds the record, but a convex reputational penalty leads experts with worse conduct to inflate more, so its effect on conduct information is non-monotone. When manipulation stakes are high, more disclosure lowers conduct information, and beyond a precision threshold even full disclosure cannot restore its peak. A record-builder indifferent to conduct over-sharpens past it, whereas unified control of both instruments avoids the problem. The Pigouvian charge for the information destroyed is zero at the threshold and so does not help. Discipline requires unified authority or an instrument pricing the crossing, such as a cap, mandate, or transfer.

Keywords: competence disclosure; signal jamming; career concerns; record manipulation; governance externality; Pigouvian correction.
JEL classification: D82; D83; G28; K13; L51.

1 Introduction

Records of expert performance are almost never governed by a single hand. A hospital system decides how finely it tracks surgical outcomes, a securities registry sets filing granularity, and a ratings platform calibrates how sharply it scores seller histories. A separate regulator then decides how much of the resulting record to publish as a competence disclosure. Health agencies mandate risk-adjusted volume statistics for physicians, and similar disclosure mandates cover financial advisers and schools.[1] The party that builds the record cares about the

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[1]New York State's risk-adjusted cardiac-surgery report cards illustrate the confounded record. FINRA's BrokerCheck and the SEC's Form ADV disclosures, layered over records built by brokerages, and state-published value-added ratings, layered over records built by schools and districts, illustrate the split between record-builder and discloser.

operational accuracy of the outcomes it tracks. The party that discloses cares about what reaches the market. Neither internalizes what its choice does to the inference problem they jointly govern. We ask what this split authority costs.

The answer turns on a mechanism we call endogenous deconfounding. A performance record confounds two persistent traits, competence and integrity, so a poor outcome is consistent with low skill, low integrity, or bad luck. Competence disclosure deconfounds it, letting the market strip out the skill component and read the remainder as evidence about conduct. But the gain is endogenous to the record technology. A sharper record raises the return to inflating it, and the expert can inflate at a convex cost. Under a convex reputational penalty the expert with more to hide gains more from looking better, so the low-integrity expert inflates more.[2] That differential inflation compresses the dimension the disclosure was meant to expose. When manipulation is strong enough, the compression overtakes the deconfounding gain, and further competence disclosure lowers rather than raises the market's information about integrity. The deconfounding the regulator buys is endogenous to the record the builder supplies.

We make this precise in a model with one expert and two policy levers. The expert has two hidden traits and produces a record that confounds them. A record-builder chooses the precision of that record, a regulator chooses the precision of a separate competence disclosure, and the market prices the expert by its posterior belief about integrity after seeing both signals. The expert privately inflates the record at a convex cost, and a convex reputational penalty makes the worse type inflate more. The two precisions are the levers, and the market's information about integrity, which we call sorting, is the object they jointly govern.

The first result is that disclosure can defeat itself. On the regular equilibrium the effect of disclosure on sorting changes sign exactly on a band of compression levels, so the comparative static is non-monotone (Theorem 1). A lever that raised the stakes alone would move sorting in one direction. The reversal is specific to the deconfounding instrument, which both sharpens a separate signal and shifts the equilibrium compression. This self-defeating band is a property of the inference problem, not yet of who controls the levers. A designer who holds both never enters it, since with a coarser record and matched disclosure it reaches the same sorting without paying the compression. The backfire is a feature of split authority, not of disclosure itself.

Split authority brings it about. Once manipulation stakes are high, a record-builder who cares only about operational accuracy sets a precision above the regulator's tolerance, and the regulator's sorting-optimal disclosure then sits at the lower edge of the self-defeating band (Theorem 2). A small direct taste for transparency pushes the equilibrium strictly inside it. The over-precision survives, under a marginal-cost condition, a builder who also values the equilibrium record, and commitment to a disclosure rule substitutes for record-margin authority only when the builder values the equilibrium record rather than the raw one (Proposition 4). The natural corrective is to price the sorting damage the record causes. But that price is zero where it is needed. The marginal sorting penalty of crossing the tolerance threshold vanishes at the crossing by an envelope identity, so the loss from crossing is second

[2] For empirical evidence of this force, see Dranove et al. (2003), who find that cardiac-surgery report cards induced provider selection responses.

order (Theorem 3); in the Gaussian-quadratic model the peak sorting value is stationary in record precision, and the zero charge then holds in closed form, while for other payoffs it requires that same peak stationarity. A sorting-damage price alone leaves the externality open. Discipline requires either unified authority or an instrument with a strictly positive charge at the threshold, a cap, mandate, or transfer that prices the crossing directly.

These pieces fix the allocation of authority. Unified control and separation paired with a record-margin instrument reach the same constrained value, while plain separation is best only when the builder's precision already falls inside the regulator's tolerance interval or when the institutional cost of intervening dominates the governance loss (Proposition 7). The two regimes also leave opposite empirical signatures. A shock that makes record technology cheaper weakly raises sorting under a single designer and lowers it under separated over-precision, so the sign of the sorting response to a record-technology shock identifies the regime (Proposition 8). The question is not whether to disclose but who must internalize the record margin.

The rest of the paper proceeds as follows. Section 2 presents the model. Section 3 derives the posterior, characterizes manipulation, and establishes the self-defeating result. Sections 4 through 7 develop the unified benchmark, the over-precision externality, the corrective failure, and the allocation of authority. Section 8 presents the application and the identifying prediction. Proofs appear in Appendices A and B. Appendices C through H contain extensions treating correlated traits, competition among record-builders, the generality of the disclosure instrument, foundations for the convex penalty, a finite-type model, and an extended application discussion.

1.1 Related literature

This paper is closest to Frankel and Kartik (2019), where a single action confounds a target trait with gaming ability and stronger stakes shift information toward gaming. We depart on three margins. The lever is the precision of a separate competence disclosure, which deconfounds the record at fixed behavior, something a stakes lever cannot do. Type dependence comes from a convex reputational penalty, so the worse type compresses the target trait's own loading. And the comparative static is non-monotone, with a self-defeating threshold, where a pure-stakes lever moves sorting monotonically.[3] A second close antecedent is Liang and Madsen (2024), who study the incentive effects of new data on an exogenous covariate. Here the record itself is manipulated, and its precision is governed separately from disclosure, so the harm runs through the interaction of two design margins.

The paper connects to several streams. First, transparency can backfire through conformism, selective withholding, receiver skepticism, or reputation dynamics under uncertain monitoring (Prat, 2005; Onuchic, 2025; Rappoport, 2025; Deb and Ishii, 2025), and records can be deleted to undo disclosed content (Pei, 2025); the correctives here act on the record-precision margin instead. Second, the disclosure lever relates to information and test design

[3] Relatedly, Kambhampati (2024) develops an anti-informativeness principle for robust performance evaluation, under which robustly optimal contracts can use joint performance evaluation even for independent agents. Here sharpening the record changes what it measures, not merely how it is used, and the harmful margin is split across two authorities. Ekmekci et al. (2022) show transparency can inhibit learning in a dynamic single-trait, single-authority model.

(Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Perez-Richet and Skreta, 2022), but sharpening the record moves the manipulation equilibrium itself, unlike verification and evidence-acquisition design (Silva, 2024; Ball and Kattwinkel, 2025; Ben-Porath et al., 2026) or competition for a receiver's belief (Gentzkow and Kamenica, 2017; Board and Lu, 2018; Grunewald and Kräkel, 2022). Third, that sharper public information can reduce welfare is known from risk-sharing and coordination (Hirshleifer, 1971; Morris and Shin, 2002; Angeletos and Pavan, 2007); here the welfare statement concerns the allocation of authority (Aghion and Tirole, 1997). Credence goods (Dulleck and Kerschbamer, 2006; Shavell, 1980; Polinsky and Shavell, 2012) and strategic classification (Hardt et al., 2016; Perdomo et al., 2020) share the manipulation incentive but not the split-authority problem.[4]

The inference structure builds on signal-jamming in career concerns (Holmström, 1999; Dewatripont et al., 1999a), with the linear reporting form of Fischer and Verrecchia (2000) and the image-concern convexity of Bénabou and Tirole (2006); Hörner and Lambert (2021) designs a motivational rating in that environment. The compression is a multitask force (Holmström and Milgrom, 1991; Dewatripont et al., 1999b) operating through inference rather than effort reallocation, and coarse evaluation can deter gaming (Ederer et al., 2018).[5]

2 Model

2.1 Primitives

A single expert has two persistent hidden traits, competence q and misalignment a, drawn independently from

$$q \sim \mathcal{N}(0, \kappa^{-1}), \qquad a \sim \mathcal{N}(0, \mu^{-1}), \qquad \kappa > 0,\ \mu > 0.$$

Higher a denotes lower integrity. The expert produces a public record

$$Y = q - a + m + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, p^{-1}), \qquad p > 0, \tag{1}$$

where ε is idiosyncratic noise and $m \in \mathbb{R}$ is hidden manipulation that inflates the record at cost

$$c(m) = \frac{m^2}{2\theta}, \qquad \theta > 0. \tag{2}$$

The parameter θ is manipulation capacity, so a larger θ lowers the marginal cost of inflating the record. The record has the signal-jamming form of the career-concerns model of Holmström (1999), with the additive reporting bias of Fischer and Verrecchia (2000). A regulator chooses the precision $\lambda \geq 0$ of a separate competence signal

$$Z = q + \nu, \qquad \nu \sim \mathcal{N}(0, \lambda^{-1}). \tag{3}$$

[4] On the record margin, a monopoly certifier optimally reveals coarsely (Lizzeri, 1999), certification precision feeds back into quality provision (Albano and Lizzeri, 2001), and consumer-score precision is itself designed (Bonatti and Cisternas, 2020); the single-designer benchmark shares the reveal-coarsely logic, and the externality arises when the certifying and disclosing hands are split. Optimal scores and decision rules underweight manipulable features (Ball, 2025; Frankel and Kartik, 2022).

[5] In our setting the coarseness that would discipline manipulation is controlled by the record-builder rather than the evaluator, which is the source of the externality.

The signal is defined for $\lambda > 0$; $\lambda = 0$ denotes no disclosure, the limit of infinite noise, and $\lambda \to \infty$ denotes perfect disclosure of competence. The noises ε and ν are independent of each other and of the traits.

The record Y is confounded. A low realization is consistent with low competence, high misalignment, or unfavorable noise. The two-trait confound is the muddled-information structure of Frankel and Kartik (2019). The disclosure Z is informative about competence only. Its role is to let the market separate the competence component of Y from its misalignment component.[6]

The market evaluates the expert through its posterior belief about misalignment,

$$M \equiv \mathbb{E}[a \mid Y, Z],$$

and pays demand

$$D(M) = d_0 - \eta M - \frac{\gamma}{2} M^2, \qquad \eta > 0,\ \gamma \geq 0. \tag{4}$$

Here $d_0 \in \mathbb{R}$ is a reputation-independent constant. The linear term is the standard reputational penalty for appearing misaligned. The quadratic term makes the penalty convex when $\gamma > 0$, so the marginal cost of an additional unit of perceived misalignment is larger for an expert who already looks worse. The image-concern form follows Bénabou and Tirole (2006). The benchmark $\gamma = 0$ recovers a penalty that is linear in the posterior.

A demand of this form has a downstream reading. Suppose receivers differ in a participation cutoff t for misalignment, with distribution F, and transact when their cutoff exceeds the public reputation, $t \geq M$. If the expert earns a fixed margin $\pi > 0$ per transaction, demand is $D(M) = \bar{d} + \pi[1 - F(M)]$, with $-D''(M) = \pi f'(M)$. The penalty is convex, $\gamma > 0$, exactly when the density of marginal receivers is increasing at the participation margin. Around a baseline $\bar{M}$, the quadratic form is the local representation with $\eta = \pi f(\bar{M})$ and $\gamma = \pi f'(\bar{M})$. What the mechanism needs is only that demand be strictly concave at the relevant reputation, $D''(\bar{M}) < 0$, so the worse type has the larger marginal gain from inflating the record. We take D as primitive. The sign $\gamma > 0$ is a substantive selection of market structures.[7]

2.2 Timing and equilibrium

The timing is as follows. The regulator commits to λ. Nature draws (q, a). The expert privately chooses m as a function of its type. The record and disclosure noises then realize, and the market observes (Y, Z), forms M, and pays $D(M)$. The expert's payoff is $D(M) - c(m)$.

[6] Restricting disclosure to a single Gaussian precision is without loss for any finite Gaussian competence experiment, and the deconfounding force extends to general Blackwell-ordered experiments; Appendix E makes both precise.

[7] The screening foundation gives $\gamma = \pi f'(\bar{M}) > 0$ when the density of marginal clients is rising at the participation margin; a mean-variance model with misalignment-scaled quality variance gives $\gamma = \pi_0 r_a c_v > 0$ exactly when risk aversion and variance-scaling are both positive; competitive assignment markets are sign-ambiguous; and the canonical multiplicative-trust model delivers negative curvature. Appendix F records the derivations.

Manipulation is not observed. The market holds a conjecture about the expert's strategy and forms M by Bayes' rule given that conjecture. We study linear equilibria in which the conjecture is confirmed.

Definition 1. A *linear signal-jamming equilibrium* is a strategy $m^*(q, a) = m_0 + m_1 a + m_2 q$ and a posterior rule $M(Y, Z)$ such that

(i) $M(Y, Z) = \mathbb{E}[a \mid Y, Z]$ under the conjecture that the expert plays m^*;

(ii) for every type (q, a), $m^*(q, a)$ maximizes $\mathbb{E}[D(M(Y, Z)) \mid q, a, m] - c(m)$ taking $M(\cdot)$ as fixed.

Because manipulation shifts the conditional mean of Y without changing its conditional variance, condition (ii) reduces to a first-order condition in m, and the linear structure of the inference problem makes the best response linear in (q, a). The market is not deceived on average. In equilibrium it subtracts the conjectured manipulation, so the martingale property $\mathbb{E}[M] = \mathbb{E}[a] = 0$ holds. Manipulation is nonetheless costly and, as Section 3.2 shows, can degrade the information content of the record.

Throughout we use the shorthand

$$K \equiv \kappa + \lambda, \qquad b \equiv 1 - m_1, \qquad \alpha \equiv 1 + m_2. \tag{5}$$

The scalar b is the equilibrium loading of the record on misalignment, as we show below, and α is its loading on competence. Compression of the misalignment signal corresponds to $b < 1$.

3 The record in equilibrium

3.1 The posterior and the discipline benefit

We derive the market's posterior and show that, holding the expert's behavior fixed, sharper competence disclosure raises the market's information about misalignment. Both results hold for any fixed linear strategy and so describe the inference environment that the expert later games.

Fix a conjectured strategy $m^c(q, a) = m_0 + m_1 a + m_2 q$ with $b = 1 - m_1$ and $\alpha = 1 + m_2$. After de-shifting by the conjectured constant, the market observes

$$Y - m_0 = \alpha q - ba + \varepsilon, \qquad Z = q + \nu. \tag{6}$$

Lemma 1. *(i) Under the conjecture (6), the posterior of (q, a) given (Y, Z) is Gaussian with precision matrix*

$$\begin{pmatrix} K + p\alpha^2 & -p\alpha b \\ -p\alpha b & \mu + pb^2 \end{pmatrix}, \tag{7}$$

where $\Delta \equiv (K + p\alpha^2)(\mu + pb^2) - p^2\alpha^2 b^2 = \mu K + p(\mu\alpha^2 + Kb^2) > 0$. The posterior mean and variance of misalignment are

$$M(Y, Z) = \frac{pb\,\{\alpha\lambda Z - K(Y - m_0)\}}{\Delta}, \qquad \mathrm{Var}(a \mid Y, Z) = \frac{K + p\alpha^2}{\Delta}, \tag{8}$$

and the cross-record variance of the posterior mean is

$$\Phi(\lambda, b, \alpha) \equiv \operatorname{Var}\big(\mathbb{E}[a \mid Y, Z]\big) = \frac{1}{\mu} - \frac{K + p\alpha^2}{\Delta} = \frac{pKb^2}{\mu\Delta}. \tag{9}$$

(ii) Holding the strategy fixed at (b, α) with $b\alpha \neq 0$,

$$\left.\frac{\partial \Phi}{\partial \lambda}\right|_{b,\alpha} = \frac{p^2 b^2 \alpha^2}{\Delta^2} > 0. \tag{10}$$

Sharper competence disclosure raises the market's information about misalignment. In any equilibrium with manipulation the loadings satisfy $b, \alpha > 0$, so the inequality is strict.

The object Φ measures how much the public record sorts experts on misalignment. By the law of total variance it equals the prior variance $1/\mu$ minus the residual posterior variance, so a larger Φ means the record is more informative about integrity. The sign convention in (8) is the expected one. Holding Z fixed, a better record lowers inferred misalignment; holding Y fixed, a stronger competence signal raises inferred misalignment, because more of a good record is attributed to competence and less to integrity.

Part (ii) is the formal content of the discipline benefit. Competence disclosure does not signal misalignment directly. It helps because the record is confounded. A sharper competence signal lets the market remove the competence component from the record and read the remainder as evidence about integrity. The residualized record

$$Y - m_0 - \frac{\alpha\lambda}{K} Z = \frac{\alpha\kappa}{K}\, q - ba + \varepsilon - \frac{\alpha\lambda}{K}\nu \tag{11}$$

makes this transparent. As λ rises, the coefficient on the competence confound q falls toward zero, so the record's residual variation increasingly reflects misalignment alone. This channel operates at fixed behavior and has no counterpart in a model where the market observes only one action and there is no separate signal to sharpen.

3.2 Manipulation and compression

We now let the expert respond. The convex penalty makes the marginal value of looking better depend on how bad the expert already looks, which ties the amount of manipulation to the trait the market is trying to learn.

Fix the market's posterior rule $M(\cdot)$ from Lemma 1(i) and consider a type (q, a) choosing actual manipulation m, on the basis of its own type and before the record and disclosure noises (ε, ν) are realized, so the strategy is a function $m(q, a)$. Because m shifts the conditional mean of Y but not its conditional variance, the expected posterior mean is

$$\bar{M}(q, a, m) \equiv \mathbb{E}[M(Y, Z) \mid q, a, m] = s_Y\, a + u\, q - s_Y(m - m_0), \qquad s_Y \equiv \frac{pbK}{\Delta}, \tag{12}$$

where $u = pb(\alpha\lambda - K)/\Delta$ is the coefficient on competence and s_Y is the sensitivity of the posterior mean to the record. The marginal benefit of manipulation is

$$\frac{\partial}{\partial m}\, \mathbb{E}[D(M) \mid q, a, m] = s_Y\big(\eta + \gamma\, \bar{M}(q, a, m)\big), \tag{13}$$

which is increasing in a when $\gamma > 0$, since $\partial \bar{M}/\partial a = s_Y > 0$ gives

$$\frac{\partial}{\partial a}\Big[s_Y\big(\eta + \gamma \bar{M}\big)\Big] = \gamma s_Y^2 > 0. \tag{14}$$

The worse the expert's expected reputation, the more a marginal unit of inflation is worth. This is the engine of type-dependent manipulation.

Proposition 1. *In any linear signal-jamming equilibrium with $\gamma > 0$,*

$$m_1 = \frac{\theta\gamma s_Y^2}{1+\theta\gamma s_Y^2} \in (0,1), \qquad b = 1 - m_1 = \frac{1}{1+\theta\gamma s_Y^2}, \tag{15}$$

and the misalignment loading of the de-shifted record (6) is $-b \in (-1,0)$, compressed from its no-manipulation value -1. The competence loading and constant satisfy

$$\alpha = \frac{bK}{\kappa+\lambda b}, \qquad m_2 = -\frac{\kappa(1-b)}{\kappa+\lambda b} < 0, \qquad m_0 = \theta\eta\, s_Y, \tag{16}$$

and b solves the scalar fixed point

$$(1-b)\,\Delta(b,\lambda)^2 = \theta\gamma\, p^2 b^3 K^2, \qquad \Delta(b,\lambda) = \mu K + p\left[\mu\left(\tfrac{bK}{\kappa+\lambda b}\right)^2 + Kb^2\right]. \tag{17}$$

When $\gamma = 0$, the unique solution is $b = 1$, so $m_1 = m_2 = 0$ and only the common shift m_0 remains. Compression strictly lowers sorting on misalignment, where the derivative is taken along the slope-consistency locus $\alpha = \alpha(b)$ with $b = 1 - m_1$,

$$\frac{d\Phi}{dm_1} < 0, \qquad \textit{hence} \qquad \Phi(\lambda, b, \alpha) < \Phi_0(\lambda) \quad \textit{whenever } b < 1, \tag{18}$$

where $\Phi_0(\lambda) = pK/\big(\mu[\mu K + p(\mu+K)]\big)$ is the no-manipulation benchmark.

The convex penalty turns manipulation into a loss of information about integrity, not merely an anticipated mean shift. A high-misalignment expert expects a worse reputation and therefore inflates more. That extra inflation flattens precisely the dimension the market is trying to read, so the misalignment loading falls from -1 to $-b$ and by (18) the record sorts experts less well on integrity.

If the posterior mean is affine in the public signals, the quadratic payoff makes the confirmed best response affine, so under the precision bound of Assumption 1 below the affine-posterior class contains a single regular equilibrium (Proposition 10). We call this branch the regular equilibrium. All subsequent results are read on it.[8]

[8] Uniqueness over unrestricted nonlinear strategies is left open; Appendix I rules out quadratic and cubic local deviations under explicit spectral and regularity conditions.

3.3 The self-defeating band

Competence disclosure raises sorting at fixed behavior (Lemma 1(ii)) but also changes equilibrium manipulation and through it the compression b. The total effect is the balance of a positive deconfounding effect and a negative compression effect. A short computation in Appendix B makes the total derivative transparent. Equilibrium sorting can be written as

$$\Phi = \frac{b\, s_Y}{\mu}, \qquad s_Y = \frac{pbK}{\Delta}, \qquad b = \frac{1}{1+\theta\gamma s_Y^2}, \tag{19}$$

so Φ depends on disclosure only through the scalar record sensitivity s_Y. Since b is a decreasing function of s_Y, the product $b\, s_Y$ need not move in the same direction as s_Y. Differentiating gives $d\Phi/d\lambda = \big[b(2b-1)/\mu\big]\, ds_Y/d\lambda$, and the sign of $ds_Y/d\lambda$ on a regular branch is the sign of $b(\lambda+2\kappa)-\kappa$. The interaction of these two factors is what allows disclosure to lower sorting.

Assumption 1 (Precision bound)**.**

$$p\Big[\mu\, \alpha_{1/2}^2 + \tfrac{K}{4}\Big] < 3\mu K, \tag{U}$$

where $\alpha_{1/2} \equiv K/(2\kappa+\lambda)$.

A primitive sufficient form, uniform in λ, is $p < 12\mu\kappa/(\kappa+\mu)$. Under Assumption 1 the linear signal-jamming equilibrium is unique and regular, with regularity denominator $\mathcal{J} \equiv 1 + 2\theta\gamma\, s_Y b^2 S_b > 0$ and $S_b = \partial s_Y/\partial b$ (Proposition 9). The comparative static below is established on that equilibrium.

Theorem 1 (Self-defeating disclosure)**.** *In the regular equilibrium, the effect of competence disclosure on sorting about misalignment is*

$$\frac{d\Phi}{d\lambda} = \frac{p^2 b^4 K^2\,(2b-1)\,\big[b(\lambda+2\kappa)-\kappa\big]}{(\kappa+\lambda b)^3\,\Delta^2\,\mathcal{J}}. \tag{20}$$

Hence

$$\frac{d\Phi}{d\lambda} < 0 \quad\Longleftrightarrow\quad \frac{\kappa}{\lambda+2\kappa} < b < \frac{1}{2} \quad\Longleftrightarrow\quad \frac{1}{2} < m_1 < \frac{\lambda+\kappa}{\lambda+2\kappa}. \tag{21}$$

On this band, additional competence disclosure lowers the market's information about misalignment.

Theorem 1 is a threshold characterization. Equation (20) holds at any regular equilibrium; (U) guarantees a unique one. The band is reached on an open set of primitives (Proposition 11). The condition $b < \frac{1}{2}$ says the worse type's inflation compresses more than half of the record's loading on misalignment; below that point the deconfounding gain dominates. The companion condition $b > \kappa/(\lambda+2\kappa)$ rules out compression so extreme that the competence-loading adjustment reverses the comparative static. Inside the band the induced compression outweighs the deconfounding gain. Figure 1 traces $b(\lambda)$ as disclosure sharpens. By contrast, under (U) equilibrium sorting is strictly decreasing in the manipulation stakes, $d\Phi/d\tau < 0$ with $\tau = \theta\gamma$. A lever that raises only the stakes moves sorting monotonically

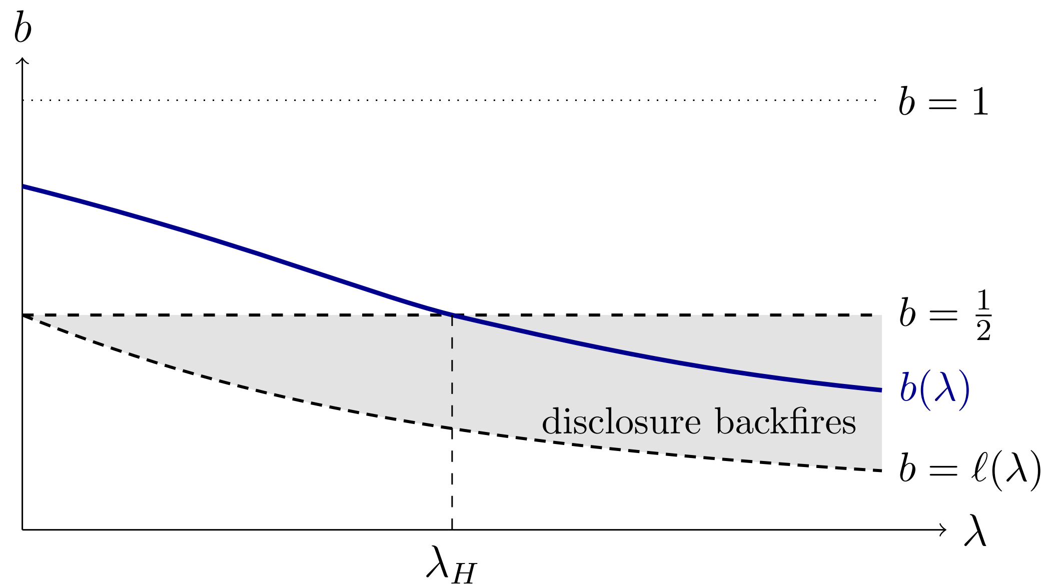

Figure 1: The self-defeating band. The equilibrium compression $b(\lambda)$ falls as competence disclosure sharpens, and disclosure backfires while it lies in the shaded band $\ell(\lambda) < b < \frac{1}{2}$, with $\ell(\lambda) = \kappa/(\lambda + 2\kappa)$. The equilibrium enters the band at λ_H, where $b = \frac{1}{2}$. The figure is schematic and shows the intermediate regime.

and cannot reproduce the non-monotone response to disclosure. The non-monotonicity is specific to the deconfounding instrument, which both sharpens a separate signal and shifts equilibrium compression, whereas a pure-stakes lever does only the latter.

Competence disclosure does two opposite things. It strips out the competence confound, making the record more revealing about integrity, and it raises the value of inflating the record, inducing the low-integrity type to flatten that informative content. When the second force dominates, sharpening the signal adds less integrity information than the induced compression removes. Write λ_H for the disclosure level at which the equilibrium first reaches $b = \frac{1}{2}$. The non-monotonicity requires both the deconfounding instrument and the convex penalty; the self-defeating threshold is the signature of their interaction.

4 Disclosure design and the unified benchmark

4.1 Optimal disclosure

Theorem 1 says competence disclosure can backfire. Proposition 2 characterizes the sorting-maximizing disclosure level for a regulator who maximizes Φ, with three cases depending on the manipulation stakes.

The compact form follows from substituting the equilibrium relation $\tau s_Y^2 = (1-b)/b$ into $\Phi = b\, s_Y/\mu$:

$$\Phi = \frac{1}{\mu}\sqrt{\frac{b(1-b)}{\tau}}, \tag{22}$$

symmetric in b about $\frac{1}{2}$ and peaking at $b = \frac{1}{2}$. Let

$$\tau_\infty \equiv \frac{4}{p^2}\Big(\mu + \tfrac{p}{4}\Big)^2, \qquad \tau_0 \equiv \frac{4}{p^2}\Big(\mu + \tfrac{p}{4} + \tfrac{p\mu}{4\kappa}\Big)^2, \tag{23}$$

with λ_H the onset level $b(\lambda_H) = \frac{1}{2}$. We maintain the primitive sufficient form of Assumption 1, $p < 12\mu\kappa/(\kappa + \mu)$, which gives (U) for every $\lambda \geq 0$.

Proposition 2. *Under Assumption 1 at every λ, in the regular equilibrium, the sorting-maximizing disclosure level is*

(i) unbounded if $\tau \leq \tau_\infty$, with Φ strictly increasing in λ, no interior maximizer, and full disclosure approached as $\lambda \to \infty$;

(ii) bounded at the backfire onset, $\lambda^* = \lambda_H$ with $b(\lambda_H) = \frac{1}{2}$, if $\tau_\infty < \tau < \tau_0$;

(iii) bounded at the lower edge, $\lambda^* = \lambda_L$ with $b(\lambda_L) = \kappa/(\lambda_L + 2\kappa)$, if $\tau > \tau_0$.

In case (ii) the optimum strictly dominates full disclosure, since the value at the onset,

$$\Phi(\lambda_H) = \frac{1}{2\mu\sqrt{\tau}}, \tag{24}$$

exceeds the full-disclosure limit $\Phi(\infty)$, the value of sorting as $\lambda \to \infty$.

In the intermediate regime (ii), disclosure should stop at the backfire onset λ_H. Figure 2 shows the single-peaked profile and the non-monotone comparison with full disclosure. Write $\Phi(0)$ and $\Phi(\infty)$ for equilibrium sorting at no disclosure and full disclosure, with equilibrium compressions b_0 and b_∞. Here $\Phi(0)$ is distinct from the no-manipulation benchmark Φ_0 of Section 3.2. On the regular branch, sorting at either extreme equals $\sqrt{b(1-b)}/(\mu\sqrt{\tau})$, so

$$\Phi(\infty) < \Phi(0) \quad \Longleftrightarrow \quad b_0 + b_\infty < 1. \tag{25}$$

There is a unique cutoff $\tau_\times \in (\tau_\infty, \tau_0)$ such that $\Phi(\infty) < \Phi(0)$ if and only if $\tau > \tau_\times$. This reversal holds throughout the strong-manipulation regime $\tau > \tau_0$. Full competence disclosure can therefore leave the market strictly less informed about integrity than no disclosure at all (Appendix B). The prescription so far values only sorting.

Lemma 2. *Gross receiver surplus $V(\lambda) = \mathbb{E}[(t - M_\lambda)_+]$, with participation cutoff $t \sim F$ independent of the type, of finite mean and nondegenerate, is a strictly increasing function of sorting $\Phi(\lambda)$. Hence $dV/d\lambda < 0$ on the self-defeating band (21), and in the intermediate regime of Proposition 2 the onset λ_H maximizes gross receiver welfare.*

Since $\mathbb{E}[M_\lambda] = 0$ and $\mathrm{Var}(M_\lambda) = \Phi(\lambda)$, a higher Φ is a mean-preserving spread of M_λ, and because $(t - M)_+$ is convex, gross receiver surplus $V(\lambda)$ rises strictly with sorting. In the intermediate regime, stopping at the onset maximizes the surplus from matching trust to integrity. If the regulator also counts manipulation costs, with welfare $w\,\Phi(\lambda) - G(\lambda)$ for a sorting weight $w > 0$ and expected equilibrium manipulation cost $G(\lambda) = \mathbb{E}[c(m^*)]$, the optimum moves weakly below the sorting optimum λ_H whenever G is nondecreasing beyond it, and no disclosure is optimal when w is small enough that $w\,\Phi' \leq G'$ below the onset.

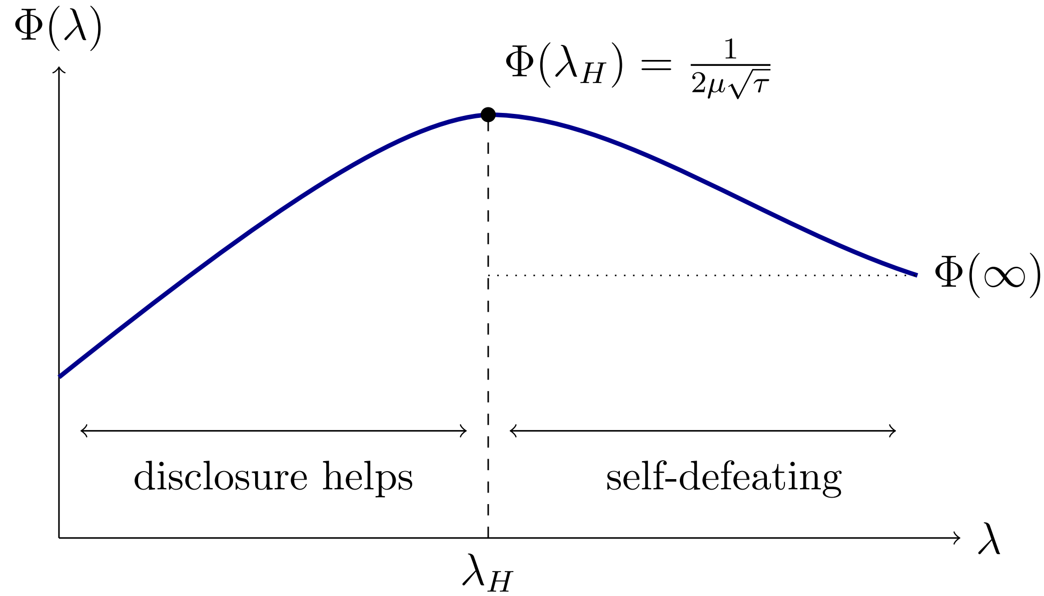

Figure 2: Information about misalignment as a function of competence-disclosure precision in the intermediate regime $\tau_\infty < \tau < \tau_0$. Sorting rises until the backfire onset λ_H, where compression reaches $b = \frac{1}{2}$, and falls thereafter; the peak $\Phi(\lambda_H) = 1/(2\mu\sqrt{\tau})$ of (24) strictly exceeds the full-disclosure limit $\Phi(\infty)$. The peak height is exact; the shape of the curve is illustrative.

Remark 1. The exact cutoffs belong to the Gaussian-quadratic specification. The mechanism rests on two open-condition signs, $\partial\Phi/\partial\lambda|_{b,\alpha} > 0$ at fixed behavior and $d\Phi/dm_1 < 0$ through type-dependent compression. Because both conditions are strict at a regular equilibrium, the sign $d\Phi/d\lambda < 0$ persists under any perturbation that preserves a nearby regular equilibrium. What the behavioral channel needs is not the global quadratic but strict monotonicity of D and strict concavity of the smoothed reputational payoff, the expectation of D over the record and disclosure noise. The zero sorting-damage charge at the threshold survives exactly the payoff classes with a stationary peak value (Theorem 3). Appendices C through I verify each piece of the mechanism beyond the Gaussian-quadratic specification.

Remark 2. When $\gamma = 0$, Proposition 1 gives $b = 1$; the market subtracts the common shift, sorting equals the no-manipulation benchmark $\Phi_0(\lambda)$ and rises monotonically with disclosure, and the band, the tolerance thresholds, and the over-precision externality all disappear. When $\gamma < 0$ the regular branch has $b > 1$, the record is amplified rather than compressed, and $d\Phi/d\lambda > 0$ by the same formula as Theorem 1, so negative curvature generates no self-defeating band.

4.2 A single designer avoids backfire

The compact form (22) makes the design problem transparent. Sorting depends on the two precisions only through the equilibrium compression b, and $\Phi = \sqrt{b(1-b)/\tau}/\mu$ is symmetric about $b = \frac{1}{2}$. A given compression $b < \frac{1}{2}$ and its mirror $1 - b > \frac{1}{2}$ deliver the same sorting, but the mirror is implemented at strictly lower record precision, because the precision that supports a compression falls as the record loads less heavily on misalignment.

Proposition 3. *Consider a designer who chooses the record precision p and the disclosure precision λ to maximize sorting Φ net of a weakly increasing record cost $C(p)$. Whenever an optimum exists, one exists with $b \geq \frac{1}{2}$, and with a strictly increasing record cost every optimum has $b \geq \frac{1}{2}$. When $\tau > \frac{1}{4}$ the maximum sorting $1/(2\mu\sqrt{\tau})$ is attained at $b = \frac{1}{2}$ along a one-parameter family of (p, λ). The self-defeating band is then never entered.*

The designer who has both instruments lowers the record's precision rather than sharpening disclosure to the point of over-manipulation, reaching the same sorting more cheaply.

The escape is not special to the Gaussian class. Under a dominance condition on the design set, requiring every over-compressed design to be matched by a weakly cheaper design on the other side of the sorting peak with weakly higher sorting, it extends to general designs. Appendix I develops the general statement, and Appendix G supplies a finite-type verification.

5 Split authority and over-precision

The record and the disclosure are rarely governed by the same party. This section characterizes what separated control costs, where the regulator's endogenous deconfounding turns counterproductive.

5.1 Separated governance and over-precision

A ratings platform, securities registry, or hospital system builds the record for its own operational purposes. The disclosure regulator observes the resulting record and chooses competence disclosure precision to maximize sorting. We model this as a sequential game in which the record-builder chooses p and the regulator observes p and chooses λ.

The record-builder values the record's information about the operational type $T = q - a$ before any competence disclosure, net of a smooth convex measurement cost C_{D}, and does not internalize the disclosure margin. The mutual information of the Gaussian record gives the value

$$B(p) = \tfrac{1}{2}\log\bigl(1 + p\,\sigma_T^2\bigr), \qquad \sigma_T^2 = \frac{1}{\kappa} + \frac{1}{\mu}, \tag{26}$$

which has a positive slope at zero that vanishes at infinity.[9] Appendix B records the microfoundation. The analysis uses the designer's value only through the scalar precision it induces, so the designer's privately optimal $p^{\mathsf{D}} > 0$ solves $B'(p^{\mathsf{D}}) = C_{\mathsf{D}}'(p^{\mathsf{D}})$, is interior, and, because the designer does not value integrity sorting, does not depend on the manipulation stakes τ.

[9] More generally, a quadratic-loss problem in which the designer estimates any payoff-relevant target X correlated with T from $Y^0 = T + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, p^{-1})$, and is paid a positive weight ζ times its risk reduction, gives $B(p) = \zeta\,\mathrm{Cov}(X,T)^2 p/(1 + p\sigma_T^2)$, strictly increasing and concave for any record-relevant target $\mathrm{Cov}(X,T) \neq 0$.

To compare regimes, let $p_0 = 4\mu/(2\sqrt{\tau} - 1)$ be the lowest record precision at which the disclosure regulator can implement $b = \frac{1}{2}$, attained as $\lambda \to \infty$, and let $p_N = 4\mu/(2\sqrt{\tau} - 1 - \mu/\kappa)$ be the highest record precision at which it can still implement $b = \frac{1}{2}$, attained at $\lambda = 0$. These satisfy $p_0 < p_N$ whenever both are positive.

Theorem 2 (Over-precision externality)**.** *Suppose manipulation stakes are high, $\tau > (1 + \mu/\kappa)^2/4$, so that p_N is positive. If the record-builder's precision exceeds p_N, then even no disclosure leaves the market over-compressed, and, on the regular separated branch (Proposition 12), the regulator's sorting-optimal disclosure sits at the lower edge of the self-defeating band, $b^R = \kappa/(\lambda^R + 2\kappa) < \frac{1}{2}$; with a small direct taste for transparency the equilibrium lies strictly inside the band. The threshold p_N is strictly decreasing in τ, from arbitrarily large near $(1 + \mu/\kappa)^2/4$ to zero as $\tau \to \infty$, so for any record-builder with an interior precision $p^{\mathsf{D}} > 0$, over-precision $p^{\mathsf{D}} > p_N$ holds if and only if*

$$\tau > \tau^*(p^{\mathsf{D}}) \equiv \frac{1}{4}\left(1 + \frac{\mu}{\kappa} + \frac{4\mu}{p^{\mathsf{D}}}\right)^2, \tag{27}$$

a cutoff strictly decreasing in p^{D}, equal to τ_0 of (23) evaluated at $p = p^{\mathsf{D}}$. Separated governance produces over-compression for every operational raw-record builder with positive interior precision once stakes pass $\tau^*(p^{\mathsf{D}})$, where a single designer would avoid it.

The proof inverts the band-edge condition to find p_N, the highest record precision at which some disclosure level can hold the market at $b = \frac{1}{2}$, then shows that any $p > p_N$ forces the regulator's sorting-optimal response to the lower band edge for every $\lambda \geq 0$. Monotonicity of p_N in τ then yields the threshold $\tau^*(p^{\mathsf{D}})$ at which the builder's fixed optimum crosses the tolerance frontier.

The conclusion uses the builder's value only through the precision it induces, so it holds for any raw-record value independent of stakes with an interior optimum. When $\mu \geq \kappa$ the regularity condition of Proposition 12 is automatic. The builder's first-order condition does not contain disclosure and is unchanged whether or not the builder anticipates the regulator's response, so over-precision follows from objectives rather than misjudgment. Any induced precision that falls more slowly than $1/\sqrt{\tau}$ still over-precises at high stakes. Three results establish that the externality is robust to the builder's objective and to the disclosure regulator's commitment power.

Proposition 4. *(i) Suppose the builder values the equilibrium record's information about the operational type $T = q - a$, scaled by $\zeta > 0$. If the initial marginal cost is low,*

$$C'_{\mathsf{D},+}(0) < \frac{\zeta}{4\kappa^2}, \tag{28}$$

then every global maximizer of the builder's problem is bounded away from zero at high stakes, and over-precision $p^{\mathsf{D}}_\tau > p_N(\tau)$ holds at all sufficiently high manipulation stakes.

(ii) Suppose the builder places weight $\omega \in [0, 1]$ on integrity sorting, with a unique interior maximizer and a strict second-order condition for each ω. If $\tau > \tau^*(p^{\mathsf{D}}(0))$, then $p^{\mathsf{D}}(\omega) > p_N$ for every $\omega \in [0, 1]$, and the welfare loss is decreasing in ω but strictly positive throughout.

(iii) Let the disclosure regulator commit, before the record is built, to any precision-contingent rule $\lambda(p)$. For raw-record builders over-precision persists under every committed rule whenever $\tau > \tau^*(p^{\mathsf{D}})$; for equilibrium-record builders a target is implementable exactly when it maximizes the builder's committed-rule objective, so commitment is ineffective under raw-record valuation and model-dependent under equilibrium-record valuation.

Condition (28) asks that the first marginal unit of precision be worth building in the high-stakes limit, where the record still loads on competence with $\alpha = \frac{1}{2}$. It is met by any cost with zero marginal cost at the origin and by small positive marginal costs. Over-precision survives every sorting weight because the marginal sorting penalty of crossing the tolerance threshold is zero at the crossing while the private marginal value of precision is strictly positive. Section 6 returns to this envelope structure. For a raw-record builder, closing the gap requires reaching the record margin directly. A cap, mandate, or transfer with a strictly positive marginal charge at the tolerance threshold accomplishes this (Appendix B).

5.2 A common objective

Whether a unified authority that also values the operational record would itself choose an over-precise record depends on the weight it places on that value. The characterization has three regions. Let the unified authority maximize $\rho\, B_{\text{raw}}(p) + \varphi_\tau(p) - C_{\mathsf{D}}(p)$, weighting operational raw-record value B_{raw} by $\rho \geq 0$ alongside sorting, where $\varphi_\tau(p) = \sup_{\lambda\geq 0} \Phi(p, \lambda; \tau)$ is the regulator's response value. For $\tau > (1+\mu/\kappa)^2/4$, the response value is flat at $\Phi^*(\tau) = 1/(2\mu\sqrt{\tau})$ on $[p_0(\tau), p_N(\tau)]$, strictly increasing below p_0, and strictly decreasing above p_N, with zero one-sided derivatives at both thresholds (Lemma 5).

Lemma 3. *Fix $\tau > (1+\mu/\kappa)^2/4$ and work on the regular branch. Let*

$$F_\rho(p) = \rho B_{\text{raw}}(p) - C_{\mathsf{D}}(p), \qquad W^I_\rho(p) = F_\rho(p) + \varphi_\tau(p).$$

Suppose B_{raw} is C^1, strictly increasing, and concave; C_{D} is C^1, strictly convex, and coercive; and F_ρ has a unique interior maximizer p^D_ρ. Let $p^I_\rho \in \arg\max_{p\geq 0} W^I_\rho(p)$. Then:

1. *If $p^D_\rho < p_0(\tau)$, every unified maximizer satisfies $p^I_\rho \in (p^D_\rho, p_0(\tau))$, and plain separation strictly lowers the common objective.*

2. *If $p^D_\rho \in [p_0(\tau), p_N(\tau)]$, then p^D_ρ is unified-optimal and plain separation creates no governance loss.*

3. *If $p^D_\rho > p_N(\tau)$, every unified maximizer satisfies $p^I_\rho \in (p_N(\tau), p^D_\rho)$, and plain separation strictly lowers the common objective.*

If, in addition, $R(p) = C'_{\mathsf{D}}(p)/B'_{\text{raw}}(p)$ is strictly increasing and the private optimum is interior for the relevant ρ, define

$$\rho_0(\tau) = R(p_0(\tau)), \qquad \rho^{\text{raw}}_N(\tau) = R(p_N(\tau)). \tag{29}$$

Then the no-loss region is equivalently $\rho \in [\rho_0(\tau), \rho^{\text{raw}}_N(\tau)]$ as defined in (29). For $\rho < \rho_0(\tau)$ separation under-precises; for $\rho > \rho^{\text{raw}}_N(\tau)$ separation over-precises.

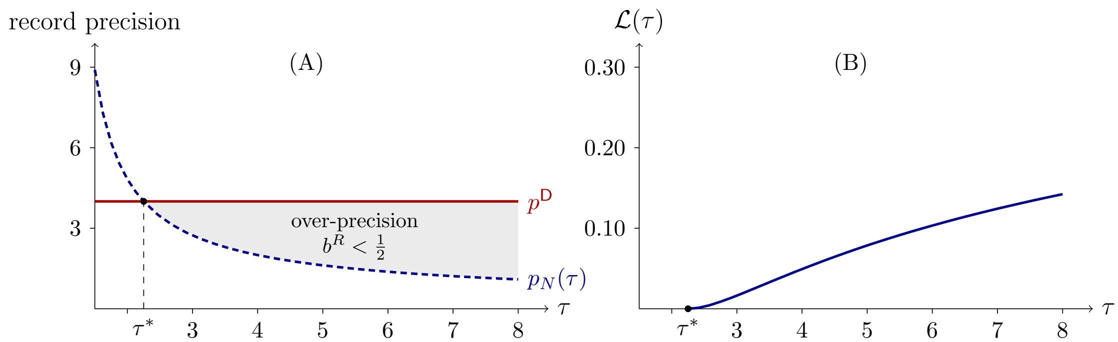

Figure 3: A computed instance of the over-precision externality. Panel A plots the builder's privately optimal precision p^{D} against the highest precision $p_N(\tau)$ at which the regulator can hold the market at $b = \frac{1}{2}$; beyond the crossing τ^* of (27) the record is over-precise and the regulator's response sits at $b^R < \frac{1}{2}$ (shaded). Panel B plots the share $\mathcal{L}(\tau)$ of attainable integrity sorting destroyed by separation, zero at τ^* and rising to 14.2% at $\tau = 8$. The configuration is that of Example 1, computed from the closed-form equilibrium.

Separated control omits the joint marginal tradeoff between operational record value and integrity sorting. When private precision exceeds $p_N(\tau)$, both regimes enter the over-compressed region, but separation enters too far. When it falls below $p_0(\tau)$, separation under-builds (Appendix B).

5.3 The welfare cost of separation

The gap between the two regimes has an exact form. At the separated equilibrium with compression b^R, sorting is $\Phi_{\text{sep}} \equiv \Phi(b^R)$ by (22), while the single designer attains $\Phi_{\text{int}} = 1/(2\mu\sqrt{\tau})$.

Lemma 4. *On the separated equilibrium the sorting loss relative to the single-designer optimum is*

$$L = \frac{1}{2\mu\sqrt{\tau}}\Big(1 - \sqrt{1 - (1 - 2b^R)^2}\Big), \tag{30}$$

strictly positive if and only if the record-builder over-precises. It is increasing in the over-precision and, holding the private precision p^{D} and the stakes τ fixed, increasing in κ and decreasing in μ. The share of attainable sorting destroyed by separation, $\mathcal{L} = L\big/\Phi_{\text{int}} = 1 - \sqrt{1 - (1 - 2b^R)^2}$, is increasing in the manipulation stakes τ when p^{D} is independent of the stakes.

The loss (30) is well defined at the band edge and measures the cost of separation wherever the record-builder over-precises. The absolute loss is not monotone in τ, because the common scale $1/\sqrt{\tau}$ shrinks both values. The governance-attributable share $\mathcal{L}$ removes that scale, and it rises with the stakes (Appendix B). Figure 3 plots the over-precision region and this loss share for a computed configuration of primitives.

The same gap reads as a receiver-surplus loss. Write the gross surplus of Lemma 2 as $V_F(\Phi) = \mathbb{E}\big[(t - \sqrt{\Phi}\,\xi)_+\big]$ with $\xi \sim \mathcal{N}(0, 1)$ and $t \sim F$. Since $V_F'(\Phi) > 0$, the separation cost in surplus units is $\Delta V = V_F(\Phi_{\text{int}}) - V_F(\Phi_{\text{sep}}) > 0$ whenever the record-builder over-precises.

Example 1. A single admissible configuration confirms the restrictions are jointly satisfiable. Take $\kappa = \mu = \gamma = \eta = \zeta = 1$, $\theta = \tau = 8$, $B(p) = \frac{1}{2}\log(1+2p)$, and $C_{\mathsf{D}}(p) = p/18 + p^2/144$, giving private precision $p^{\mathsf{D}} = 4$. This clears (U), the strong-manipulation regime ($\tau = 8 > \tau_0 = 2.25$), over-precision ($p_N(8) = 1.094 < 4$), separated lower-edge regularity, and (28) ($C'_{\mathsf{D},+}(0) = 1/18 < 1/4$). The regulator responds with $\lambda^R = 2.12$, leaving $b^R = 0.243$ and $\Phi_{\text{sep}} = 0.152$ against the single-designer value $\Phi_{\text{int}} = 0.177$, a loss $L = 0.025$ and share $\mathcal{L} = 14.2\%$. For every $p^{\mathsf{D}} > 0$ the externality operates on the unbounded open interval $\tau > \tau^*(p^{\mathsf{D}})$, and the governance-attributable share $\mathcal{L}$ grows with τ (Lemma 4), so the deeper into the over-precision region the economy is pushed, the larger the fraction of attainable sorting destroyed.

Boundary facts delimit the externality. The mechanism requires a record that loads jointly on competence and misalignment. Without that joint structure, the condition $b\alpha \neq 0$ of Lemma 1(ii) fails and there is nothing to deconfound. Separation creates no governance loss when the builder's precision falls within the tolerance interval $[p_0(\tau), p_N(\tau)]$, or equivalently when the operational weight ρ lies in the no-loss band $[\rho_0(\tau), \rho_N^{\text{raw}}(\tau)]$ of Lemma 3. The threshold structure extends to records of any fixed competence-integrity composition, with p_N moving continuously in the composition (Appendix E).

6 Constrained efficiency and the corrective failure

The sorting loss of Section 5 is a constrained-efficiency loss in a stated surplus sense, and the textbook Pigouvian response fails to close it at the tolerance threshold. Let total surplus be

$$W(p, \lambda) = B(p) - C_{\mathsf{D}}(p) + v\, V_F(\Phi(p, \lambda)) - \varsigma\, G(p, \lambda),$$

where $v > 0$ is the receiver weight (the surplus-unit analogue of the sorting weight w of Section 4.1) and $\varsigma \geq 0$ is the manipulation-deadweight weight, $V_F(\Phi) = \mathbb{E}[(t - \sqrt{\Phi}\,\xi)_+]$ is the receiver surplus of Lemma 2, and $G(p, \lambda) = \mathbb{E}[c(m^*(p, \lambda))]$ is the expected equilibrium manipulation cost. Define the reduced objects $\varphi(p) = \Phi(p, \lambda^R(p; \tau))$ and $g(p) = G(p, \lambda^R(p; \tau))$ after composing with the sorting regulator's best response λ^R. Gross receiver surplus rises with sorting by construction, since $(t - M)_+$ is convex and a higher Φ is a mean-preserving spread of the posterior. The welfare content of what follows is therefore not that monotonicity but the comparison across governance regimes, where the operational value B and the manipulation deadweight G enter with their own weights and the sorting-damage price fails at the threshold. The record-margin constrained-efficient precision p^{C} maximizes

$$U_0(p) + v\, V_F(\varphi(p)) - \varsigma\, g(p),$$

over $p \geq 0$, with $U_0(p) = B(p) - C_{\mathsf{D}}(p)$. This benchmark fixes disclosure authority with the sorting regulator and asks only what a planner would choose on the record margin.

The separated builder, who maximizes $U_0(p)$ alone, omits two external costs. The first is the marginal receiver-sorting damage from raising precision; the second, weighted by ς, is the marginal manipulation deadweight. On the over-precision branch both have the same sign, and their sum defines the governance wedge $\mathcal{W}(p) = -v\, V_F'(\varphi(p))\varphi'(p) + \varsigma\, g'(p)$. The following proposition records the constrained-inefficiency of separation.

Proposition 5. *Fix τ and a regulator response $\lambda^R(p;\tau)$. Let $\varphi(p) = \Phi(p, \lambda^R(p;\tau))$ and $g(p) = G(p, \lambda^R(p;\tau))$, and define*

$$S(p) = U_0(p) + v\, V_F(\varphi(p)) - \varsigma\, g(p), \qquad U_0(p) = B(p) - C_{\mathsf{D}}(p),$$

with $v > 0$ and $\varsigma \geq 0$. Suppose $V_F' > 0$, φ and g are differentiable on the over-precision branch, and on that branch $\varphi'(p) < 0$ and $g'(p) > 0$. If the separated builder's choice p^{D} is an interior maximizer of U_0 and $p^{\mathsf{D}} > p_N(\tau)$, then the separated allocation is not record-margin constrained-efficient. In particular, for all sufficiently small $\varepsilon > 0$,

$$S(p^{\mathsf{D}} - \varepsilon) > S(p^{\mathsf{D}}).$$

The wedge $\mathcal{W}(p)$ is strictly positive on the over-precision branch. The sorting component is positive because $V_F' > 0$ and $\varphi' < 0$; the deadweight component $\varsigma\, g'(p)$ is positive when $\varsigma > 0$. Hence $S'(p^{\mathsf{D}}) = -\mathcal{W}(p^{\mathsf{D}}) < 0$, confirming the local improvement. When $\varsigma = 0$ this reduction of the full surplus problem to the record margin is exact regardless of the disclosure response, because the planner's disclosure first-order condition then coincides with the sorting regulator's. When $\varsigma > 0$ a planner controlling both margins would raise disclosure above the sorting response on the over-precision branch, so the reduced benchmark is conservative there. Appendix I develops the full two-margin benchmark. A mandate $p = p^{\mathsf{C}}$ implements p^{C} directly, and a cap $p \leq p^{\mathsf{C}}$ does so whenever U_0 is strictly increasing on $[0, p^{\mathsf{C}}]$. The substance lies in how the threshold crossing is priced. The following proposition characterizes the instruments that implement p^{C} and the complementary corrective on the conduct margin.

Proposition 6. *Let $p^{\mathsf{C}} \in \arg\max_p S(p)$, where*

$$S(p) = U_0(p) + v\, V_F(\varphi(p)) - \varsigma\, g(p).$$

(i) The transfer

$$T^{\mathsf{P}}(p) = -v\, V_F(\varphi(p)) + \varsigma\, g(p) + t_0$$

implements the constrained-efficient argmax set. If p^{C} is the unique global maximizer of S, the transfer implements p^{C} exactly. Where differentiable, its marginal charge is

$$(T^{\mathsf{P}})'(p) = -v\, V_F'(\varphi(p))\varphi'(p) + \varsigma\, g'(p) = \mathcal{W}(p).$$

(ii) A quadratic liability $r(m) = \frac{\rho_L}{2} m^2$ on manipulation acts as a reduction of the manipulation capacity from θ to $\theta_L \equiv (\theta^{-1} + \rho_L)^{-1} < \theta$. Under Assumption 1, holding disclosure fixed, stronger liability lowers the inflation slope m_1, raises the misalignment loading b, and raises sorting on misalignment, $d\Phi/d\rho_L > 0$.

A cap $p \leq p_N$ prevents the builder from entering the region where the regulator can no longer attain the sorting peak, but is constrained-efficient only when $S(p_N) \geq S(p)$ for all p. A sorting-damage-only charge cannot hold the builder at p_N when $U_0'(p_N) > 0$, the precise sense in which the natural Pigouvian price fails. The full Pigouvian marginal charge at the threshold is $(T^{\mathsf{P}})'_+(p_N) = \varsigma\, g'_+(p_N)$, positive only when $\varsigma > 0$. It implements p_N only when p_N is itself constrained-efficient, requiring locally $\varsigma\, g'_+(p_N) \geq U_0'(p_N)$. The constrained-efficient precision p^{C} may lie below p_N (when sorting gains dominate), equal p_N (when the deadweight component is large enough), or exceed p_N (when $\varsigma = 0$). The mandate, the cap, and the transfer all implement the reduced benchmark, in which disclosure authority remains with the sorting regulator. Theorem 3 states the envelope structure behind this failure.

The vanishing sorting-damage charge is an envelope property whose reach depends on peak-value stationarity. Let equilibrium sorting be $\psi(x, p)$ for scalar compression state x and record precision p near p_N. For each p near p_N there is a unique interior peak $x^*(p)$ with $\psi_x = 0$ and $\psi_{xx} < 0$, and peak value $V(p) \equiv \psi(x^*(p), p)$. The threshold p_N is the largest precision at which the disclosure response implements $x^*(p)$. On the implementable side $\varphi(p) = V(p)$, and the over-precision response branch $x^R(p)$ satisfies $x^R(p_N) = x^*(p_N)$ with bounded one-sided derivative at p_N.

Theorem 3 (Zero charge at the threshold)**.** *Under the three hypotheses above,*

$$\varphi'_+(p_N) = V'_+(p_N).$$

In particular, $\varphi'_+(p_N) = 0$ if and only if the peak value is locally stationary at the threshold, $V'_+(p_N) = 0$. If V is locally constant to first order and the peak is nondegenerate, the loss from crossing the threshold is second order, with leading coefficient

$$\frac{-\psi_{xx}(x^*(p_N), p_N)}{2} \left((x^R)'_+(p_N) - (x^*)'_+(p_N) \right)^2.$$

At p_N, the over-precision response touches the interior sorting peak, so movement of the constrained branch has no first-order effect on sorting. The right derivative equals the derivative of the peak value, $\varphi'_+(p_N) = V'_+(p_N)$. The Gaussian-quadratic model sets this to zero in closed form, since the peak value $1/(2\mu\sqrt{\tau})$ is constant on the tolerance interval. For non-quadratic reputational payoffs the same zero-charge conclusion requires a stationary peak value. If smoothed curvature varies so that the peak value changes at first order, the sorting-only marginal price $-v\, V'_F(\varphi(p_N))\, V'_+(p_N)$ is nonzero.

Part (ii) treats the conduct directly rather than the record margin. Because the harm in Theorem 1 is caused by disclosure-induced manipulation, liability raises the cost of manipulation without acting on either precision margin. The liability raises the effective manipulation cost from θ^{-1} to θ_L^{-1}, lowering the inflation slope m_1, compression, and by $d\Phi/dm_1 < 0$, raising sorting. Inside the self-defeating band liability and disclosure move sorting in opposite directions, so liability has the favorable effect there. The manipulation first-order condition becomes

$$c'(m) + r'(m) = s_Y\big(\eta + \gamma \bar{M}(q, a, m)\big). \tag{31}$$

Alternative signal-based correctives underweight manipulable features (Ball, 2025; Frankel and Kartik, 2022). Underweighting discards the record variation that, once deconfounded, reveals integrity, whereas liability lowers the inflation slope while leaving both signals intact.

7 The allocation of authority

This section asks which structure an authority allocator would choose. Let $\mathcal{G} = \{\mathsf{U}, \mathsf{S}, \mathsf{S}^{\mathsf{R}}, \mathsf{S}^{\lambda}\}$, where U unifies authority, S separates without an instrument, S^{R} separates with a precommitted record-margin cap, mandate, or transfer, and S^{λ} separates with only a precommitted disclosure rule. The allocator maximizes $\rho\, B_{\text{raw}}(p) + \Phi(p, \lambda; \tau) - C_{\mathsf{D}}(p)$ net of fixed institutional costs $\iota_g \geq 0$. Write V_U and V_S for the common surplus under U and S.

Proposition 7. *Let $\Delta_S(\rho, \tau) = V_U(\rho, \tau) - V_S(\rho, \tau)$, where V_U is the common-objective value under unified authority and V_S the value under plain separation.*

(i) U *and* S^{R} *attain the same constrained common value whenever the record-margin instrument in* S^{R} *implements the unified target p^I_ρ. The target is p^I_ρ, not generically p_0 or p_N; it lies in (p^D_ρ, p_0) in the under-precision region, equals p^D_ρ in the no-loss band, and lies in (p_N, p^D_ρ) under over-precision.*

(ii) *For raw-record builders,* S^λ *is value-equivalent to* S*, because disclosure-rule commitment alone does not change the builder's raw-record objective. Let $\iota_I = \min\{\iota_{\mathsf{U}}, \iota_{\mathsf{S}^{\mathsf{R}}}\}$ and $\iota_P = \min\{\iota_{\mathsf{S}}, \iota_{\mathsf{S}^\lambda}\}$. Then a unified-value implementing structure is optimal iff $\Delta_S(\rho, \tau) \geq \iota_I - \iota_P$, and a plain or disclosure-only separated structure is optimal iff $\Delta_S(\rho, \tau) \leq \iota_I - \iota_P$. In the no-loss band $p^D_\rho \in [p_0(\tau), p_N(\tau)]$, $\Delta_S = 0$ and the allocator chooses the cheapest structure in $\mathcal{G}$. Outside that band, $\Delta_S > 0$, and plain separation is chosen only when its fixed-cost advantage exceeds the common-value loss.*

The optimal assignment is margin-specific. Plain separation is dominated only when the builder's precision falls outside the tolerance interval $[p_0(\tau), p_N(\tau)]$, and a committed disclosure rule substitutes for record-margin authority only for equilibrium-record builders, and then only when the rule makes the unified target globally incentive compatible (Appendix B).

8 Application and identification

The mechanism applies to settings where a competence or quality score is published for experts whose records confound skill and conduct, with one institution building the record and a second disclosing competence on top of it. Physician scorecards and credentialed financial advisers are the two lead cases. In each, a record-builder (hospital system, brokerage) sets precision and a regulator mandates competence disclosure. The curvature condition $\gamma > 0$ is most natural when marginal clients have increasing density at the retention cutoff or when perceived conflicts raise quality variance for risk-averse clients, and is not automatic where multiplicative trust governs demand. The two governance regimes leave opposite signatures in how integrity sorting responds to record technology, formalized through a cost shifter χ that raises the marginal cost of record precision.

Proposition 8. *Suppose the shifter χ raises the marginal cost of record precision, the record optima are interior and strict, the separated response is on the regular lower-edge branch, and the unified benchmark is the sorting-maximizing designer. Then under single-designer governance cheaper record technology weakly raises integrity sorting, $d\Phi_{\text{int}}/d\chi \leq 0$, while under separated governance with over-precision it lowers integrity sorting, $d\Phi_{\text{sep}}/d\chi > 0$. The sign of the sorting response to a record-technology shock therefore differs across the two regimes.*

Under separation, cheaper record technology induces more precision, deepening over-compression and lowering sorting; under unified governance the same shock weakly helps, because the designer internalizes the sorting margin. The reversal requires an endogenous record-design margin held by a party distinct from the disclosure regulator; when diagnosticity is exogenous, a technology shock has no channel. The diagnostic requires the sorting-maximizing unified benchmark.[10] Scorecards, school ratings, and online seller scores differ in whether the record-builder and the discloser are the same hand, so the comparison is feasible across observed regimes.

9 Conclusion

In many settings, performance records that confound competence and integrity are built by one institution and disclosed by another. We have shown that this division of authority is costly. Once manipulation stakes are high, a record-builder that does not value integrity sorting sharpens the record past the regulator's tolerance, and the externality survives a builder that also values the equilibrium record and any disclosure rule the regulator commits to in advance. Past that threshold no disclosure level restores the sorting peak, and the market settles below the integrity information a single designer would attain. When the builder's precision does not respond to the stakes, the share of attainable sorting that separation destroys grows with the temptation to manipulate, whereas a single designer holding both instruments never enters this region.

The natural corrective fails exactly where it is needed. The sorting-damage component of the Pigouvian charge vanishes at the tolerance threshold by an envelope property, exactly when the peak sorting value is stationary in record precision as it is here, while the full charge that also prices manipulation deadweight stays positive. Closing the externality therefore requires unified authority or a record-margin instrument with a strictly positive charge at the crossing, such as a cap, mandate, or transfer. Plain separation is enough only when the builder's precision falls within the regulator's tolerance interval or when the cost of intervening exceeds the governance loss. The prescription is margin-specific. The question is not whether to disclose but who must internalize the record margin. The cost traces to endogenous deconfounding, the deconfounding the regulator buys being endogenous to the record the builder supplies, and its signature is a sign reversal in how integrity sorting responds to record technology that a fixed-confounding model does not produce.

Future research should examine richer design games where record and disclosure precisions are chosen under partial information and conflicting objectives. Competition among record-builders is not intrinsically corrective. Below the reward-weight threshold $\omega^*_{\mathsf{P}}(\tau)$ rivalry amplifies over-precision; above it rivalry disciplines precision, and in the pure sorting-reward case prevents over-precision altogether (Appendix D). A dynamic model would connect the over-investment channel here to the deletion channel of Pei (2025).

[10] The sign divergence is observable when the governance regime is observed, demand is strictly monotone on the relevant support, so that the reputation M and its cross-sectional dispersion are recovered from observed valuations, and the record-technology shifter moves record precision without directly moving demand, the priors, or what is manipulable.

A Equilibrium uniqueness and the self-defeating band

The fixed point (17) can be written $T(b) = \theta\gamma$ with $T(b) \equiv (1-b)\Delta(b,\lambda)^2/(p^2b^3K^2)$, which falls from $+\infty$ at $b=0$ to 0 at $b=1$, so the equilibrium is unique when T is decreasing.

Proposition 9. *Under Assumption 1, for every $\theta\gamma > 0$ the fixed point (17) has a unique root $b \in (0,1)$, and that root satisfies the regularity condition $\mathcal{J} > 0$ of Theorem 1.*

Proof of Proposition 9. Write $T(b) = (1-b)\Delta(b)^2/(p^2b^3K^2)$ with $\Delta(b) = \mu K + pA(b)$ and $A(b) = \mu\alpha(b)^2 + Kb^2$, where $\alpha(b) = bK/(\kappa + \lambda b)$, so the fixed point (17) is $T(b) = \theta\gamma$. Logarithmic differentiation gives

$$\frac{bT'(b)}{T(b)} = -\frac{b}{1-b} - 3 + 2b\frac{\Delta'(b)}{\Delta(b)}.$$

Since $b\alpha'(b)/\alpha(b) = \kappa/(\kappa+\lambda b) \le 1$, we have $bA'(b) \le 2A(b)$, hence $b\Delta'(b)/\Delta(b) = pbA'(b)/(\mu K + pA) \le 2pA/(\mu K + pA) < 2$. For $b \ge \frac{1}{2}$ this already gives $bT'/T < -b/(1-b) - 3 + 4 \le 0$. For $b < \frac{1}{2}$, monotonicity of α and b^2 gives $A(b) < A_{1/2} \equiv \mu\alpha_{1/2}^2 + K/4$, and condition (U), namely $pA_{1/2} < 3\mu K$, gives $pA(b)/(\mu K + pA) < \frac{3}{4}$, so $b\Delta'/\Delta < \frac{3}{2}$ and $bT'/T < -b/(1-b) - 3 + 3 < 0$. Thus $T' < 0$ on $(0,1)$, and since T falls from $+\infty$ to 0 there is a unique root. At a root, $\theta\gamma\, s_Y^2 = (1-b)/b$, and a direct computation gives $\mathcal{J} = -b(1-b)T'(b)/T(b) > 0$. □

Assumption 1 bounds the precision of the record relative to the priors, so that the endogenous response of the posterior sensitivity does not overturn the single crossing of the manipulation fixed point. It restricts information precision, not the intensity of manipulation. Under Assumption 1 the slope-consistency relation $\alpha = bK/(\kappa + \lambda b)$ makes the map $b \mapsto (m_0, m_1, m_2)$ a bijection onto linear equilibria, so the unique root is the unique linear equilibrium and (21) is read on it with no selection.

Multiplicity when the bound fails. The regular roots ($\mathcal{J} > 0$, equivalently $T'(b) < 0$) are the stable rest points of the adjustment $\dot{b} = \mathcal{B}(b) - b$; the branch from $\theta\gamma = 0$, $b = 1$ is regular, and (21) holds on it. Proposition 10 below gives uniqueness within the affine-posterior class. The affine class is closed, so the regular branch is the only equilibrium with affine posterior beliefs.

Proposition 10. *Consider any regular Bayes-confirmed equilibrium in which the posterior mean $M(y,z) = \mathbb{E}[a \mid Y = y, Z = z]$ is affine in the public signals (y,z). Then the expert's confirmed best response is affine in (q,a). Hence, under Assumption 1, the equilibrium coincides with the unique regular affine signal-jamming equilibrium.*

Proof of Proposition 10. Write $M(y,z) = \beta_0 + \beta_Y y + \beta_Z z$. For a type choosing record level $t = q - a + m$, the smoothed reputational payoff is $A(t,q) = \mathbb{E}[D(M(t+\varepsilon, q+\nu))]$. Since D is quadratic and (ε,ν) are Gaussian and independent of (t,q), A is a quadratic polynomial in (t,q) with $A_t(t,q) = c_0 + c_t t + c_q q$ and $c_t = -\gamma\beta_Y^2 \le 0$. The best-response condition $m/\theta = A_t(q - a + m, q)$ is linear in m with unique maximizer

$$m(q,a) = \frac{\theta c_0 + \theta(c_t + c_q)q - \theta c_t\, a}{1 + \theta\gamma\beta_Y^2},$$

affine in (q, a). Bayes confirmation forces the conjectured strategy to equal this affine rule. The induced record law is Gaussian and the posterior is the affine Gaussian posterior of the main text, so the equilibrium is a root of (17). Under (U) that root is unique. $\square$

A full local uniqueness theorem over all regular measurable equilibria would require a uniform spectral bound on the linearized best response at every polynomial degree, together with a C^1 mapping theorem for $\mathcal{B}$ on a Hermite–Sobolev space; the uniform bound remains open. The self-defeating band of Theorem 1 is reached on an open set of primitives.

Proposition 11. *Suppose Assumption 1 holds and $\lambda > 0$. Then the equilibrium satisfies $\frac{1}{2} < m_1 < (\lambda+\kappa)/(\lambda+2\kappa)$, and hence $d\Phi/d\lambda < 0$, if and only if*

$$T_{1/2} < \theta\gamma < T_\ell, \tag{SD}$$

where, with $\ell \equiv \kappa/(\lambda+2\kappa)$,

$$T_{1/2} = \frac{4\Delta_{1/2}^2}{p^2K^2}, \qquad T_\ell = \frac{(1-\ell)\Delta_\ell^2}{p^2\ell^3K^2},$$

$\Delta_{1/2} = \mu K + p\big[\mu\alpha_{1/2}^2 + \frac{K}{4}\big]$, and $\Delta_\ell = \mu K + p\big[\frac{\mu}{4} + K\ell^2\big]$. The interval (SD) is nonempty whenever $\lambda > 0$.

Proof of Proposition 11. Under condition (U), T is strictly decreasing, so the unique root satisfies $b < \frac{1}{2}$ if and only if $\theta\gamma > T(\frac{1}{2})$ and $b > \ell$ if and only if $\theta\gamma < T(\ell)$. Evaluating T at the two points, with $\alpha(\frac{1}{2}) = \alpha_{1/2} = K/(2\kappa+\lambda)$ and $\alpha(\ell) = \frac{1}{2}$, gives $T(\frac{1}{2}) = 4\Delta_{1/2}^2/(p^2K^2) = T_{1/2}$ and $T(\ell) = (1-\ell)\Delta_\ell^2/(p^2\ell^3K^2) = T_\ell$, where $\Delta_{1/2}$ and Δ_ℓ are as stated in the proposition. Because $\ell < \frac{1}{2}$ for $\lambda > 0$ and T is decreasing, $T_\ell > T_{1/2}$, so condition (SD) is a nonempty open interval. The equivalence with $\frac{1}{2} < m_1 < (\lambda+\kappa)/(\lambda+2\kappa)$ follows from $b = 1 - m_1$, and the sign of $d\Phi/d\lambda$ then follows from Theorem 1. $\square$

The lower bound, $m_1 > \frac{1}{2}$, requires the worse type's inflation to compress more than half the record's loading on misalignment. The upper bound asks that manipulation not be so extreme that the competence-loading adjustment $m_2 < 0$ reverses the comparative static. Disclosure backfires in between. Figure 1 traces $b(\lambda)$ as disclosure sharpens.

Because (U) must hold along the entire fixed-point map it is conservative. The separated equilibrium sits at a single response λ^R, admitting a sharper characterization. Write $y^R = 1 - 2b^R$ and $\bar{m} = \mu/\kappa$.

Proposition 12. *At the separated lower-edge equilibrium $b^R = \kappa/(\lambda^R + 2\kappa)$, the regularity denominator satisfies $\mathcal{J} > 0$ if and only if*

$$\sqrt{\tau} > G_{\bar{m}}(y^R) := \frac{(1+y^R)\sqrt{1-(y^R)^2} + \bar{m}/\sqrt{1-(y^R)^2}}{2+y^R}.$$

This region strictly contains the region certified by the uniform bound (U). If $\mu \geq \kappa$, every feasible lower-edge point is regular, with no cap on p^{D}. If $\mu < \kappa$, regularity can fail only for $0 < y^R < \sqrt{1-\mu/\kappa}$ and $\sqrt{\tau} \leq G_{\bar{m}}(y^R)$, that is, at low stakes.

Proof of Proposition 12. Log-differentiating the fixed point $T(b) = (1-b)\Delta(b,\lambda)^2/(p^2b^3K^2)$ gives $T'/T = -1/(1-b) - 3/b + 2\Delta_b/\Delta$, so $\mathcal{J} = -b(1-b)T'/T = 3 - 2b - 2b(1-b)\Delta_b/\Delta$. Evaluating at the lower edge b^R, where $\alpha = \frac{1}{2}$ and $s_Y = \sqrt{(1-b^R)/(\tau b^R)}$, and writing $w = \sqrt{1-(y^R)^2}$ and $s_\tau = \sqrt{\tau}$, the denominator reduces to $\mathcal{J}_\partial = 2 + y^R - [(1+y^R)w + \bar{m}/w]/s_\tau$, so $\mathcal{J} > 0$ if and only if $s_\tau > G_{\bar{m}}(y^R)$. The lower-edge precision map is feasible if and only if $s_\tau > F_{\bar{m}}(y^R)$ with $F_{\bar{m}}(y) = \frac{1}{2}(w + \bar{m}/w)$, and $G_{\bar{m}}(y) - F_{\bar{m}}(y) = y(w^2 - \bar{m})/[2w(2+y)]$. If $\bar{m} \geq 1$, then $w^2 - \bar{m} \leq 0$ for all $y \in (0,1)$, so $G_{\bar{m}} \leq F_{\bar{m}}$ and feasibility implies regularity, with $p^{\mathsf{D}} = 4\mu w/[(1-y)^2(2s_\tau - w - \bar{m}/w)]$ unbounded as the feasibility denominator vanishes. If $\bar{m} < 1$, irregularity requires $F_{\bar{m}} < s_\tau \leq G_{\bar{m}}$, hence $w^2 > \bar{m}$, that is $y^R < \sqrt{1-\bar{m}}$. At the onset $y^R = 0$, $\mathcal{J}_\partial = 2 - (1+\bar{m})/s_\tau > 0$ whenever $s_\tau > (1+\bar{m})/2$, while condition (U) is compatible with over-precision only for $s_\tau > 2(1+\bar{m})/3$, so the interval $(1+\bar{m})/2 < s_\tau \leq 2(1+\bar{m})/3$ is regular yet outside condition (U). $\square$

The separated comparative static of Theorem 2 is stated for this regular region.

B Proofs

B.1 Proof of Lemma 1

Part (i). The posterior precision matrix is the prior precision plus the sum of signal precisions times their loading outer products,

$$\operatorname{diag}(\kappa,\mu) + p\begin{pmatrix}\alpha \\ -b\end{pmatrix}\begin{pmatrix}\alpha & -b\end{pmatrix} + \lambda\begin{pmatrix}1 \\ 0\end{pmatrix}\begin{pmatrix}1 & 0\end{pmatrix} = \begin{pmatrix}K + p\alpha^2 & -p\alpha b \\ -p\alpha b & \mu + pb^2\end{pmatrix},$$

using $K = \kappa+\lambda$. Its determinant is $\Delta = (K+p\alpha^2)(\mu+pb^2) - p^2\alpha^2b^2 = \mu K + p(\mu\alpha^2 + Kb^2) > 0$. The posterior covariance is the matrix inverse, so $\operatorname{Var}(a \mid Y, Z) = (K+p\alpha^2)/\Delta$. The posterior mean solves the precision-weighted system; its second component is $M = pb\{\alpha\lambda Z - K(Y - m_0)\}/\Delta$. By the law of total variance $\Phi = 1/\mu - (K+p\alpha^2)/\Delta = pKb^2/(\mu\Delta)$.

Part (ii). Differentiate $\Phi = pKb^2/(\mu\Delta)$ with respect to λ holding (b,α) fixed. Since $\partial K/\partial\lambda = 1$ and $\partial\Delta/\partial\lambda|_{b,\alpha} = \mu + pb^2$,

$$\left.\frac{\partial\Phi}{\partial\lambda}\right|_{b,\alpha} = \frac{pb^2}{\mu}\cdot\frac{\Delta - K(\mu+pb^2)}{\Delta^2} = \frac{pb^2}{\mu}\cdot\frac{p\mu\alpha^2}{\Delta^2} = \frac{p^2b^2\alpha^2}{\Delta^2} > 0,$$

where the middle equality uses $\Delta - K(\mu+pb^2) = p\mu\alpha^2$. $\square$

B.2 Proof of Proposition 1

The proof has three steps. The first derives the best-response condition from the quadratic payoff and matches coefficients to recover m_0, m_1, and m_2. The second imposes slope consistency between the strategy and the posterior to obtain the scalar fixed point in b. The third signs the effect of compression on sorting by differentiating Φ along the slope-consistency locus.

Marginal benefit and best response. Since manipulation shifts the conditional mean of Y without affecting its conditional variance, $\partial\mathbb{E}[D(M) \mid q, a, m]/\partial m = s_Y(\eta + \gamma\bar{M})$ by (12) and $\partial\bar{M}/\partial m = -s_Y$. The first-order condition $m/\theta = s_Y(\eta + \gamma\bar{M}(q, a, m))$ is linear in m; matching coefficients against $m^c = m_0 + m_1 a + m_2 q$ gives $m_0 = \theta\eta s_Y$, $m_1 = \theta\gamma s_Y^2/(1+\theta\gamma s_Y^2)$, and $m_2 = m_1 u/s_Y$. Strict concavity holds with curvature $-\gamma s_Y^2 - 1/\theta < 0$.

Slope consistency and the fixed point. Since $u = pb(\alpha\lambda - K)/\Delta$, imposing $\alpha = 1 + m_2$ gives $\alpha = bK/(\kappa + \lambda b)$ and $m_2 = -\kappa(1-b)/(\kappa + \lambda b) < 0$. Writing $\theta\gamma s_Y^2 = (1-b)/b$ and substituting $s_Y = pbK/\Delta$ yields the scalar fixed point $(1-b)\Delta^2 = \theta\gamma p^2 b^3 K^2$. When $\gamma = 0$ the right-hand side vanishes and $b = 1$; existence for $\gamma > 0$ follows from continuity.

Compression lowers sorting. Along $\alpha = bK/(\kappa + \lambda b)$,

$$\frac{d}{db}\,\Phi(\lambda, b) = \frac{2pKb}{\Delta^2}\left[K + \frac{p\lambda b^3 K^2}{(\kappa + \lambda b)^3}\right] > 0,$$

and $b = 1 - m_1$, so $d\Phi/dm_1 < 0$ and $b < 1$ implies $\Phi < \Phi_0(\lambda)$. $\square$

B.3 Proof of Theorem 1

The proof has four steps. The first expresses equilibrium sorting as a function of the record sensitivity s_Y and signs its derivative. The second differentiates s_Y along the equilibrium branch to obtain $ds_Y/d\lambda$. The third combines these to produce the total derivative (20) and characterizes its sign. The fourth shows that a stakes lever moves sorting monotonically on the regular branch.

Sorting as a function of record sensitivity. From $\Phi = b\, s_Y/\mu$ and $\theta\gamma s_Y^2 = (1-b)/b$, differentiating and substituting gives

$$\frac{d\Phi}{ds_Y} = \frac{1}{\mu}\Big(b + s_Y\frac{db}{ds_Y}\Big) = \frac{b(2b-1)}{\mu}.$$

Branch derivative of record sensitivity. Along the equilibrium branch $b = 1/(1 + \theta\gamma S^2)$, implicit differentiation in λ gives $ds_Y/d\lambda = S_\lambda/\mathcal{J}$ with $\mathcal{J} = 1 + 2\theta\gamma s_Y b^2 S_b > 0$ on a regular branch and

$$S_\lambda = \frac{\mu p^2 b^3 K^2\,[\,b(\lambda + 2\kappa) - \kappa\,]}{(\kappa + \lambda b)^3\,\Delta^2}.$$

Total derivative and threshold. Combining:

$$\frac{d\Phi}{d\lambda} = \frac{b(2b-1)}{\mu}\cdot\frac{S_\lambda}{\mathcal{J}} = \frac{p^2 b^4 K^2\,(2b-1)\,[\,b(\lambda + 2\kappa) - \kappa\,]}{(\kappa + \lambda b)^3\,\Delta^2\,\mathcal{J}},$$

which is (20). The sign equals that of $(2b-1)[b(\lambda + 2\kappa) - \kappa]$; since $\kappa/(\lambda + 2\kappa) \le \frac{1}{2}$ for $\lambda \ge 0$, this is negative exactly on $\kappa/(\lambda + 2\kappa) < b < \frac{1}{2}$. Translating through $b = 1 - m_1$ gives (21).

The stakes lever. Along the branch $T(b) = \tau$ with $T(b) = (1-b)\Delta^2/(p^2 b^3 K^2)$, implicit differentiation gives $db/d\tau = -b(1-b)/(\mathcal{J}\tau) < 0$. From the compact form $\Phi = \sqrt{b(1-b)/\tau}/\mu$,

$$\frac{d\Phi}{d\tau} = -\frac{\Phi}{2\mathcal{J}\tau}(\mathcal{J} + 1 - 2b), \qquad \mathcal{J} + 1 - 2b = 2(1-b)\Big(2 - \frac{b\Delta_b}{\Delta}\Big).$$

Using $\alpha - b\alpha_b = \lambda b^2 K/(\kappa + \lambda b)^2 \geq 0$, a direct computation gives $2\Delta - b\Delta_b = 2\mu[K + p\lambda b^3 K^2/(\kappa + \lambda b)^3] > 0$, so $\mathcal{J} + 1 - 2b > 0$ and $d\Phi/d\tau < 0$ on the regular branch. □

B.4 Proof of Proposition 2

By Theorem 1, $\operatorname{sgn} \Phi'(\lambda) = \operatorname{sgn}\{(2b-1)(b-\ell)\}$ with $\ell = \kappa/(\lambda + 2\kappa) \leq \frac{1}{2}$.

Value at the onset. At $b = \frac{1}{2}$, the fixed point (17) gives $\tau = 4\Delta^2/(p^2K^2)$, so $\Delta/K = \frac{p}{2}\sqrt{\tau}$ and $\Phi(\lambda_H) = pK(\frac{1}{2})^2/(\mu\Delta) = 1/(2\mu\sqrt{\tau})$, confirming (24).

Full-disclosure limit. As $\lambda \to \infty$, $\alpha \to 1$ and $\Delta/K \to \mu + pb^2$, so b_∞ solves $(1 - b_\infty)(\mu + pb_\infty^2)^2 = \tau p^2 b_\infty^3$ and $\Phi(\infty) = pb_\infty^2/(\mu(\mu + pb_\infty^2))$. Substituting τ from the fixed point into $\Phi(\lambda_H)$,

$$\Phi(\lambda_H) - \Phi(\infty) = \frac{pb_\infty^{3/2}}{2\mu(\mu + pb_\infty^2)\sqrt{1 - b_\infty}}\Big[1 - 2\sqrt{b_\infty(1 - b_\infty)}\Big] \geq 0,$$

since $b(1-b) \leq \frac{1}{4}$, with equality only at $b_\infty = \frac{1}{2}$.

Onset map and cases. At $b = \frac{1}{2}$ the fixed-point map is $T(\frac{1}{2}; \lambda) = 4\Delta_{1/2}^2/(p^2K^2)$. Since $\alpha_{1/2}^2/K = (\kappa + \lambda)/(\lambda + 2\kappa)^2$ is strictly decreasing in λ, the onset map falls strictly from τ_0 at $\lambda = 0$ to τ_∞ as $\lambda \to \infty$. In case (i) $\tau \leq \tau_\infty$, so $b(\lambda) > \frac{1}{2}$ throughout and Φ is increasing with no interior maximum. In case (ii) $\tau_\infty < \tau < \tau_0$, yielding a unique λ_H with $b(\lambda_H) = \frac{1}{2}$; the interior peak dominates full disclosure by the display above, so $\lambda^* = \lambda_H$. In case (iii) $\tau > \tau_0$, so $b(0) < \frac{1}{2}$; Φ rises to the lower edge and falls thereafter, giving $\lambda^* = \lambda_L$ with $b(\lambda_L) = \ell(\lambda_L)$.

The transparency reversal. At any equilibrium $\Phi = \sqrt{b(1-b)}/(\mu\sqrt{\tau})$, so $\Phi(\infty) < \Phi(0)$ iff $b_\infty(1 - b_\infty) < b_0(1 - b_0)$. Since $x(1-x)$ is symmetric about $\frac{1}{2}$, this is equivalent to $b_0 + b_\infty < 1$. Write the endpoint fixed point as $T_c(b) = (1-b)(\mu + pcb^2)^2/(p^2b^3) = \tau$, where $c = 1$ at full disclosure and $c_0 = 1 + \mu/\kappa$ at no disclosure. Under condition (U), T_c is strictly decreasing in b and increasing in c, so $b_0 > b_\infty$. Implicit differentiation gives $db_c/d\tau < 0$ at each endpoint, so the sum $\mathcal{S}(\tau) = b_0 + b_\infty$ is strictly decreasing. Since $\mathcal{S}(\tau_\infty) > 1$ (with $b_\infty = \frac{1}{2}$) and $\mathcal{S}(\tau_0) < 1$ (with $b_0 = \frac{1}{2}$), a unique $\tau_\times \in (\tau_\infty, \tau_0)$ solves $\mathcal{S} = 1$. Setting $b_\infty = x_\times$, $b_0 = 1 - x_\times$ and equating $T_1(x_\times) = T_{c_0}(1 - x_\times)$, using $c_0 - 1 = \mu/\kappa$, yields $1 - 2x_\times = (p/\kappa)x_\times^2(1 - x_\times)^2$; the left side is decreasing and the right is increasing on $(0, \frac{1}{2})$, so $x_\times$ is unique. □

B.5 Proof of Lemma 2

Since $\mathbb{E}[M_\lambda] = 0$ and $\mathrm{Var}(M_\lambda) = \Phi(\lambda)$, a higher Φ is a mean-preserving spread of the Gaussian M_λ. Because $(t - M)_+$ is convex in M, Jensen's inequality gives $V(\lambda) = \mathbb{E}[(t - M_\lambda)_+]$ strictly increasing in $\Phi(\lambda)$. The sign statements then follow from Theorem 1 and Proposition 2. $\square$

B.6 Proof of Proposition 3

By (22), sorting depends on (p, λ) only through b, with $\Phi(b) = \varrho(b)/\mu$ and $\varrho(b) = \sqrt{b(1-b)/\tau}$ symmetric about $\frac{1}{2}$. Write $P(b, \lambda)$ for the record precision supporting compression b at disclosure λ, the unique positive solution of (17) for p, and $P_\infty(b) \equiv \lim_{\lambda \to \infty} P(b, \lambda)$. Fix any feasible design (b, λ) with $b < \frac{1}{2}$ and consider its mirror $1-b$; symmetry gives $\Phi(1-b) = \Phi(b)$. In the full-disclosure limit $P_\infty(b) = \mu\varrho(b)/\{b^2[1-\varrho(b)]\}$ and $P_\infty(1-b)/P_\infty(b) = b^2/(1-b)^2 < 1$, so since $P(b, \lambda) \geq P_\infty(b) > P_\infty(1-b)$, the mirror reaches the same sorting at strictly lower record precision. With a strictly increasing record cost every $b < \frac{1}{2}$ design is strictly dominated by its mirror, so no optimum has $b < \frac{1}{2}$. Under a weakly increasing cost the mirror is weakly cheaper, so whenever an optimum exists an optimum with $b \geq \frac{1}{2}$ exists. Without a record cost the same sorting is available at $b = \frac{1}{2}$ whenever $\tau > \frac{1}{4}$, with supporting values tracing a one-parameter family. $\square$

B.7 Proof of Theorem 2

The proof proceeds in two parts. The first establishes that any record precision $p > p_N$ leaves the regulator's sorting-optimal response at the lower band edge for every disclosure level, and verifies the monotone dependence of p_N on τ. The second solves for the threshold $\tau^*(p^{\mathsf{D}})$ at which the builder's privately optimal precision crosses p_N.

Fix $\tau > (1 + \mu/\kappa)^2/4$, so that $p_N = 4\mu/(2\sqrt{\tau} - 1 - \mu/\kappa)$ is positive and exceeds $p_0 = 4\mu/(2\sqrt{\tau} - 1)$, the two denominators differing by $\mu/\kappa > 0$. By construction p_0 is the lowest record precision implementing $b = \frac{1}{2}$ as $\lambda \to \infty$, and p_N the highest implementing $b = \frac{1}{2}$ at $\lambda = 0$, so $p_0 < p_N < \infty$. Suppose the record-builder's precision satisfies $p > p_N$. Since p_N is the largest precision compatible with $b = \frac{1}{2}$ even at the no-disclosure boundary $\lambda = 0$, no admissible $\lambda \geq 0$ restores $b = \frac{1}{2}$, and the market is over-compressed with $b < \frac{1}{2}$ for every $\lambda \geq 0$. By (22) sorting rises toward its peak at $b = \frac{1}{2}$, so the regulator maximizes sorting by choosing the smallest feasible compression, attained at the lower band edge $b^R = \ell(\lambda^R) = \kappa/(\lambda^R + 2\kappa) < \frac{1}{2}$, with $p = P(b^R, \lambda^R)$ and $d\Phi/d\lambda = 0$. By Theorem 1, on $\ell(\lambda) < b < \frac{1}{2}$ one has $d\Phi/d\lambda < 0$, so the sorting-only optimum sits exactly at the lower edge; a direct transparency taste $\varpi H(\lambda)$ with $H' > 0$ and $\varpi > 0$ small shifts the first-order condition to $d\Phi/d\lambda = -\varpi H'(\lambda^R) < 0$, placing the perturbed equilibrium strictly inside the band. For the private precision, the designer maximizes $B(p) - C_{\mathsf{D}}(p)$ with B increasing and concave. The mutual-information value $B(p) = \frac{1}{2}\log(1 + p\sigma_T^2)$ has $B'(p) = \frac{1}{2}\sigma_T^2/(1 + p\sigma_T^2)$, falling from $\frac{1}{2}\sigma_T^2$ to zero; the quadratic-loss value is the Bayes-risk reduction $\sigma_T^2 - \mathrm{Var}(T \mid Y^0) = \sigma_T^4 p/(1 + p\sigma_T^2)$, using $\mathrm{Var}(T \mid Y^0) = (\sigma_T^{-2} + p)^{-1}$, with derivative $\sigma_T^4/(1 + p\sigma_T^2)^2$ falling from σ_T^4 to zero. For a general payoff-relevant target X correlated with T, the same computation gives $\mathrm{Var}(X) - \mathrm{Var}(X \mid Y^0) = \mathrm{Cov}(X, T)^2 p/(1 + p\sigma_T^2)$, of which $X = T$ is the displayed case.

For either value, under a smooth convex C_{D} with $C'_{\mathsf{D}}(0^+) < B'(0^+)$ and slope eventually exceeding B', the strictly concave objective has a unique interior maximizer $p^{\mathsf{D}} > 0$ solving $B'(p^{\mathsf{D}}) = C'_{\mathsf{D}}(p^{\mathsf{D}})$, which carries no τ.

Since $p_N(\tau) = 4\mu/(2\sqrt{\tau} - 1 - \mu/\kappa)$ falls from ∞ to 0 as τ rises from $(1 + \mu/\kappa)^2/4$ to ∞, there is a unique τ^* with $p_N(\tau^*) = p^{\mathsf{D}}$. Solving $2\sqrt{\tau^*} - 1 - \mu/\kappa = 4\mu/p^{\mathsf{D}}$ gives $\tau^*(p^{\mathsf{D}}) = \frac{1}{4}(1 + \mu/\kappa + 4\mu/p^{\mathsf{D}})^2$. A single designer implements $b = \frac{1}{2}$ at $p_0 < p_N$ using the full-disclosure boundary. □

B.8 Formal complements and proof of Proposition 4

The equilibrium-record value function is

$$\widetilde{B}(p, \lambda; \tau) = \zeta \, \frac{(\alpha/\kappa + b/\mu)^2}{\alpha^2/\kappa + b^2/\mu + 1/p}, \tag{32}$$

where (α, b) are the regular affine loadings and $\zeta > 0$ is a value scale. Write $B^R(p; \tau) = \widetilde{B}(p, \lambda^R(p; \tau); \tau)$ with $B^R(0; \tau) = 0$, let $B_\infty(p) = \zeta \, p/(\kappa(4\kappa + p))$ be the high-stakes limit of B^R, and let $V_\infty = \sup_{p \geq 0}\{B_\infty(p) - C_{\mathsf{D}}(p)\}$.

The subsection proves the three parts of the proposition. Part (i) derives the high-stakes limit of the builder's equilibrium-record objective and shows over-precision holds whenever that limit has positive value. Part (ii) shows over-precision persists under any sorting weight by the zero derivative of φ at p_N. Part (iii) addresses committed disclosure rules.

Part (i). Write the equilibrium-record value as $\widetilde{B} = \zeta \mathcal{N}^2/V$ with $\mathcal{N} = \alpha/\kappa + b/\mu$ and $V = \alpha^2/\kappa + b^2/\mu + 1/p$, from $\mathrm{Cov}(T, Y - m_0) = \alpha/\kappa + b/\mu$ and $\mathrm{Var}(Y - m_0) = \alpha^2/\kappa + b^2/\mu + 1/p$ for $T = q - a$.

Local shape at p_N. Two branches of the response meet at p_N, where $\alpha = b = \frac{1}{2}$. On the peak branch $b = \frac{1}{2}$, parameterizing by $\alpha \in [\frac{1}{2}, 1]$,

$$P_H(\alpha) = \frac{\mu\kappa}{\kappa(\sqrt{\tau}/2 - 1/4) - \mu\alpha(1 - \alpha)}, \qquad P_H(1) = p_0, \quad P_H(\tfrac{1}{2}) = p_N, \quad P'_H(\tfrac{1}{2}) = 0,$$

so $d\alpha/dp \to -\infty$ as $p \uparrow p_N$; since $\widetilde{B}_\alpha > 0$, the left derivative is $(B^R)'_-(p_N) = -\infty$. On the lower-edge branch $\alpha = \frac{1}{2}$, with $y = 1 - 2b$ and $w = \sqrt{1 - y^2}$,

$$\lambda^R = \frac{2\kappa y}{1 - y}, \qquad P_L(y) = \frac{4\mu w}{(1 - y)^2(2\sqrt{\tau} - w - \bar{m}/w)}, \qquad P_L(0) = p_N, \quad P'_L(0) = 2p_N,$$

with $\bar{m} = \mu/\kappa$, so $b_p(p_N^+) = -1/(4p_N)$. Combining $\widetilde{B}_p$ and $\widetilde{B}_b \, b_p$ at $\alpha = b = \frac{1}{2}$ gives

$$(B^R)'_+(p_N) = \zeta \, \frac{\mathcal{N}_N}{2\kappa \, p_N^2 V_N^2} > 0, \qquad \mathcal{N}_N = \tfrac{1}{2}\big(\tfrac{1}{\kappa} + \tfrac{1}{\mu}\big), \quad V_N = \tfrac{1}{4\kappa} + \tfrac{1}{4\mu} + \tfrac{1}{p_N}.$$

High-stakes limit and over-precision. For fixed $p > 0$, as $\tau \to \infty$ the lower-edge response has $b^R(p;\tau) = O(\tau^{-1/3})$ and $\alpha^R = \frac{1}{2}$, so

$$B^R(p;\tau) \to B_\infty(p) = \zeta\,\frac{(1/(2\kappa))^2}{1/(4\kappa) + 1/p} = \zeta\,\frac{p}{\kappa(4\kappa+p)}, \qquad B'_\infty(0) = \frac{\zeta}{4\kappa^2}.$$

Since B_∞ is strictly increasing and concave and C_{D} is convex and coercive, $B_\infty - C_{\mathsf{D}}$ is strictly concave with right derivative $\zeta/(4\kappa^2) - C'_{\mathsf{D},+}(0)$ at the origin, so $V_\infty > 0$ if and only if $C'_{\mathsf{D},+}(0) < \zeta/(4\kappa^2)$, which is condition (28). When $V_\infty > 0$ the unique limiting maximizer is positive, and uniform convergence of $B^R(\cdot;\tau)$ to B_∞ on compact subsets away from the vanishing boundary layer, together with $B^R(p;\tau) = O(p_N(\tau))$ on $[0, p_N]$, gives by argmax continuity a $\delta > 0$ with every global maximizer $p^{\mathsf{D}}_\tau > \delta$ for all high τ; since $p_N(\tau) \to 0$, $p^{\mathsf{D}}_\tau > p_N(\tau)$ eventually. Conversely, when $V_\infty = 0$ every sequence of maximizers converges to 0, so none is bounded away from zero.

Boundary layer. When $V_\infty = 0$ the high-stakes optimum collapses to zero, but literal over-precision can persist in the boundary layer where both p^{D}_τ and $p_N(\tau)$ vanish; condition (28) governs the positive-limit case.

Part (ii). Write $U_0(p) = B(p) - C_{\mathsf{D}}(p)$, $\varphi(p) = \Phi(b^R(p))$, and $W_\omega(p) = U_0(p) + \omega\varphi(p)$. At $p = p_N$ the regulator implements $b^R = \frac{1}{2}$, so $\Phi_b(\frac{1}{2}) = 0$ and $\varphi_p(p_N^+) = \Phi_b(\frac{1}{2})\, b^R_p(p_N^+) = 0$. Strict concavity of U_0 with interior maximum $p^{\mathsf{D}}(0) > p_N$ gives $U'_0(p_N) > 0$. Hence $W_{\omega,p}(p_N^+) = U'_0(p_N) > 0$ for every $\omega \in [0,1]$, so p_N is never a maximizer and $p^{\mathsf{D}}(\omega) > p_N$. On the over-precision branch $\Phi_b(b^R) > 0$ and $b^R_p < 0$, so $\varphi_p < 0$. The implicit-function rule gives $\partial p^{\mathsf{D}}/\partial\omega = -\varphi_p/W_{\omega,pp} < 0$, and $dL/d\omega < 0$ while $b^R(p^{\mathsf{D}}(\omega)) < \frac{1}{2}$ keeps $L > 0$.

Part (iii). The builder maximizes $U_0(p) = B(p) - C_{\mathsf{D}}(p)$, which does not contain λ. Under any committed rule $\lambda(p)$ the interior optimum solves $B'(p) = C'_{\mathsf{D}}(p)$ and equals p^{D}; over-precision $p^{\mathsf{D}} > p_N(\tau)$ then holds whenever $\tau > \tau^*(p^{\mathsf{D}})$ by Theorem 2. A transfer priced at the marginal sorting damage adds $\varphi(p)$ to the objective, which is the $\omega = 1$ case of part (ii); since $\varphi_p(p_N^+) = 0$ while $U'_0(p_N) > 0$, the builder still crosses p_N. For equilibrium-record builders the committed rule enters the objective and a target $\hat{p}$ is implementable iff $\widetilde{B}(\hat{p}, \lambda(\hat{p}); \tau) - C_{\mathsf{D}}(\hat{p}) \geq \widetilde{B}(p, \lambda(p); \tau) - C_{\mathsf{D}}(p)$ holds for every $p \geq 0$. $\square$

B.9 The response-value lemma

Lemma 5 (Response value). *Fix $\tau > (1+\mu/\kappa)^2/4$ and let $\Phi^*(\tau) = 1/(2\mu\sqrt{\tau})$. The disclosure regulator's response value*

$$\varphi_\tau(p) = \sup_{\lambda \geq 0} \Phi(p, \lambda; \tau)$$

is continuous on $(0, \infty)$ and satisfies:

(i) Strictly increasing on $(0, p_0(\tau))$, where $p_0(\tau) = 4\mu/(2\sqrt{\tau} - 1)$.

(ii) Constant at $\Phi^*(\tau)$ on $[p_0(\tau), p_N(\tau)]$, where $p_N(\tau) = 4\mu/(2\sqrt{\tau} - 1 - \mu/\kappa)$.

(iii) Strictly decreasing on $(p_N(\tau), \infty)$.

(iv) Zero one-sided derivatives at both thresholds:

$$\varphi'_{\tau,-}(p_0) = \varphi'_{\tau,+}(p_0) = 0, \qquad \varphi'_{\tau,-}(p_N) = \varphi'_{\tau,+}(p_N) = 0.$$

The lemma characterizes the disclosure regulator's response value in four steps. The first covers the under-precision branch where the supremum is approached as $\lambda \to \infty$. The second covers the plateau where $b = \frac{1}{2}$ is implementable. The third covers the over-precision branch where the best response sits at the lower edge. The fourth establishes zero one-sided derivatives at both thresholds via the chain rule and the vanishing of Φ_b at $b = \frac{1}{2}$.

The sorting index satisfies $\Phi_b(b) = (1-2b)/(2\mu\sqrt{\tau}\sqrt{b(1-b)})$, positive for $b < 1/2$, zero at $b = 1/2$, and negative for $b > 1/2$.

Under-precision branch ($p < p_0$). The full-disclosure limit compression $b_\infty(p) > 1/2$ solves $(1-b)(\mu+pb^2)^2 = \tau p^2 b^3$, with $b'_\infty(p) < 0$ by implicit differentiation. Since $\Phi_b(b_\infty) < 0$, $\varphi'_\tau(p) = \Phi_b(b_\infty)b'_\infty(p) > 0$; the supremum is approached as $\lambda \to \infty$.

Plateau ($p \in [p_0, p_N]$). The regulator can implement $b = 1/2$, attaining $\Phi^*(\tau)$, so $\varphi'_\tau = 0$ on the plateau.

Over-precision branch ($p > p_N$). The best response sits at the lower edge with $b_L < 1/2$ and $b'_L(p) < 0$, so $\varphi'_\tau(p) = \Phi_b(b_L)b'_L(p) < 0$.

Zero one-sided derivatives. At each threshold $b = 1/2$, so $\Phi_b(1/2) = 0$ and the chain rule gives zero. The quadratic contact $\varphi_\tau(p) = \Phi^*(\tau) - (p-p_j)^2/(16\mu\sqrt{\tau}\,p_j^2) + o((p-p_j)^2)$ near $p_j \in \{p_0, p_N\}$ confirms that active branches touch the plateau tangentially. □

B.10 Proof of Lemma 3

By Lemma 5, φ_τ is strictly increasing on $[0, p_0)$, flat at $\Phi^*(\tau) = 1/(2\mu\sqrt{\tau})$ on $[p_0, p_N]$, and strictly decreasing on (p_N, ∞), with zero one-sided derivatives at p_0 and p_N. Since F_ρ is strictly concave with unique interior maximizer p_ρ^D, it is strictly increasing to p_ρ^D and strictly decreasing thereafter. If $p_\rho^D \in [p_0, p_N]$, then $F_\rho(p) + \varphi_\tau(p) \le F_\rho(p_\rho^D) + \Phi^*(\tau) = F_\rho(p_\rho^D) + \varphi_\tau(p_\rho^D)$ for all p, so the separated precision is also unified-optimal and there is no governance loss.

If $p_\rho^D < p_0$, then $(W_\rho^I)'(p_\rho^D) = F'_\rho(p_\rho^D) + \varphi'_\tau(p_\rho^D) = \varphi'_\tau(p_\rho^D) > 0$, so increasing precision strictly raises the common objective. Points below p_ρ^D are dominated because both F_ρ and φ_τ are no larger there; points above p_0 are dominated because F_ρ is decreasing there and $\varphi'_\tau(p_0^-) = 0$. Hence every unified maximizer lies in (p_ρ^D, p_0).

If $p_\rho^D > p_N$, then $(W_\rho^I)'(p_\rho^D) = \varphi'_\tau(p_\rho^D) < 0$, so lowering precision strictly raises the common objective. Points above p_ρ^D are dominated because both F_ρ and φ_τ are no larger there; points below p_N are dominated because F_ρ is increasing there and $\varphi'_\tau(p_N^+) = 0$. Hence every unified maximizer lies in (p_N, p_ρ^D). If $R(p) = C'_{\mathsf{D}}(p)/B'_{\mathrm{raw}}(p)$ is strictly increasing, the private first-order condition $R(p_\rho^D) = \rho$ makes p_ρ^D strictly increasing in ρ, giving the stated thresholds. □

B.11 Proof of Lemma 4

Write $y = 1 - 2b^R \in [0, 1)$ and $w = \sqrt{1-y^2} = 2\sqrt{b^R(1-b^R)}$. On the single-designer optimum $b = \frac{1}{2}$, $\Phi_{\text{int}} = 1/(2\mu\sqrt{\tau})$. The sorting loss is

$$L = \Phi_{\text{int}} - \Phi(b^R) = \frac{1}{\mu\sqrt{\tau}}\left(\tfrac{1}{2} - \sqrt{b^R(1-b^R)}\right) = \frac{1-w}{2\mu\sqrt{\tau}}.$$

$L > 0$ iff $b^R < \frac{1}{2}$, and $\partial L/\partial y = y/(2\mu\sqrt{\tau}w) > 0$, so L increases in over-precision. Differentiating the edge map confirms $\partial L/\partial\kappa > 0$ and $\partial L/\partial\mu < 0$. The share $\mathcal{L} = L/\Phi_{\text{int}} = 1 - w = 1 - \sqrt{1-(1-2b^R)^2}$ is free of $1/\sqrt{\tau}$. Since p^{D} solves $B' = C'_{\mathsf{D}}$ independently of τ, implicit differentiation along the lower-edge branch gives $dy/d\tau > 0$, so $\mathcal{L}$ is strictly increasing in τ. □

B.12 Proof of Proposition 5

$U_0'(p^{\mathsf{D}}) = 0$ at the builder's optimum. On the over-precision branch $b^R < \frac{1}{2}$ and $b_p^R < 0$, so $\varphi'(p) = \Phi_b(b^R)\, b_p^R < 0$, making the sorting component $-vV_F'(\varphi)\varphi' > 0$. At the lower edge $\alpha = \frac{1}{2}$, $m_1 = 1 - b$, $m_2 = -1/2$, and writing the deadweight along the lower edge as a function of b, $g_b(b) = -\eta^2/(2\gamma b^2) - (1-b)/(\theta\mu) < 0$; since $b_p^R < 0$, $g'(p) > 0$ and the deadweight component $\varsigma g' > 0$ when $\varsigma > 0$. Hence

$$S'(p^{\mathsf{D}}) = U_0'(p^{\mathsf{D}}) + vV_F'(\varphi)\varphi' - \varsigma g' = -\mathcal{W}(p^{\mathsf{D}}) < 0,$$

and $S(p^{\mathsf{D}} - \varepsilon) > S(p^{\mathsf{D}})$ for small $\varepsilon > 0$. □

B.13 Proof of Proposition 6

This proposition has two parts. Part (i) characterizes the Pigouvian transfer on the record margin. Part (ii) characterizes liability on the conduct margin.

Part (i): transfer. With the charge $T^{\mathsf{P}}(p) = -vV_F(\varphi(p)) + \varsigma g(p) + t_0$, the builder maximizes $U_0(p) - T^{\mathsf{P}}(p) = S(p) - t_0$, so maximizers coincide with $\arg\max S$. At p_N, $\varphi'_+(p_N) = 0$. So the sorting-damage-only marginal price is

$$-vV_F'(\varphi(p_N))\,\varphi'_+(p_N) = 0.$$

A builder with $U_0'(p_N) > 0$ facing only this charge has positive marginal payoff at p_N and crosses. The full Pigouvian marginal charge is

$$(T^{\mathsf{P}})'_+(p_N) = \varsigma\, g'_+(p_N), \qquad g'_+(p_N) = \frac{1}{4p_N}\left(\frac{2\eta^2}{\gamma} + \frac{1}{2\theta\mu}\right) > 0.$$

The full charge is positive iff $\varsigma > 0$, and implements p_N locally iff $\varsigma g'_+(p_N) \geq U_0'(p_N)$.

Part (ii): liability. With $r(m) = \frac{\rho_L}{2}m^2$ the manipulation first-order condition reads $m(\frac{1}{\theta} + \rho_L) = s_Y(\eta + \gamma\bar{M})$, which is the condition of Proposition 1 with θ replaced by $\theta_L = (\theta^{-1} + \rho_L)^{-1} < \theta$. The fixed point becomes $T(b) = \theta_L\gamma$. Under (U), T is strictly decreasing, so raising ρ_L lowers θ_L, raises b, and lowers $m_1 = 1 - b$. Since $\partial\Phi/\partial b > 0$ (from the proof of Proposition 1), $d\Phi/d\rho_L > 0$ at fixed λ. □

B.14 Proof of Theorem 3

Differentiating $\varphi(p) = \psi(x^R(p), p)$ at p_N^+ and using $\psi_x(x^*(p_N), p_N) = 0$ (interior peak) with bounded $(x^R)'_+(p_N)$ gives $\varphi'_+(p_N) = \psi_p(x^*(p_N), p_N) = V'_+(p_N)$. When V is locally constant the second-order expansion yields the stated leading coefficient. $\square$

Gaussian specialization. The branch slope is $(x^R)'_+(p_N) = -1/(4p_N)$. The peak slope is $(x^*)'_+(p_N) = 0$. Also, $-\psi_{xx} = 2/(\mu\sqrt{\tau})$.

The leading coefficient is $1/(16\mu\sqrt{\tau}\, p_N^2)$, matching Lemma 5.

The zero-charge conclusion requires $V'_+(p_N) = 0$; single-peakedness alone does not deliver it.

In the Gaussian-quadratic model the peak value $1/(2\mu\sqrt{\tau})$ is constant, so the condition holds in closed form. Appendix I characterizes the non-quadratic payoff classes with a stationary peak value.

B.15 Proof of Proposition 7

Part (i): equivalence. The common objective depends on λ only through Φ. At any fixed p, a sorting-maximizing regulator chooses $\lambda^R(p;\tau) = \arg\max_\lambda \Phi$, so delegating disclosure does not reduce the common objective. If the record-margin instrument implements p_ρ^I (located by Lemma 3: in (p_ρ^D, p_0) under under-precision, at p_ρ^D in the no-loss band, and in (p_N, p_ρ^D) under over-precision), the induced allocation attains the same common value.

Part (ii): raw-record builders. Proposition 4(iii) gives that any committed disclosure rule leaves the builder's objective unchanged, so $\mathsf{S}^\lambda \equiv \mathsf{S}$. In the no-loss band $\Delta_S = 0$ and all structures are value-equivalent; outside it $\Delta_S > 0$ and an implementing structure is preferred iff $\Delta_S \geq \iota_I - \iota_P$. $\square$

B.16 Proof of Proposition 8

Let $\Phi_b(b) = (1-2b)/(2\mu\sqrt{\tau b(1-b)})$, positive for $b < \frac{1}{2}$ and non-positive for $b \geq \frac{1}{2}$. By Proposition 3, the single designer sits at $b_C \geq \frac{1}{2}$. Writing its record-precision first-order condition as $\mathcal{F}(b_C, \chi) = 0$ with $\mathcal{F}_b < 0$ at a strict optimum, and using $\mathcal{F}_\chi = -(\partial C_p/\partial\chi)P'_\infty(b_C) > 0$ (the shifter raises marginal cost and $P'_\infty(b_C) < 0$), implicit differentiation gives $db_C/d\chi > 0$. Since $\Phi_b(b_C) \leq 0$, we have $d\Phi_{\text{int}}/d\chi = \Phi_b(b_C)\, db_C/d\chi \leq 0$.

By Theorem 2, the separated regulator sits at $b^R < \frac{1}{2}$. The platform's choice satisfies $B'(p^{\mathsf{D}}) = C'_{\mathsf{D}}(p^{\mathsf{D}};\chi)$, so $dp^{\mathsf{D}}/d\chi < 0$. On the regular lower-edge branch $\partial b^R/\partial p^{\mathsf{D}} < 0$, so $db^R/d\chi > 0$.

Since $\Phi_b(b^R) > 0$, it follows that

$$d\Phi_{\text{sep}}/d\chi = \Phi_b(b^R)\, db^R/d\chi > 0.$$

$\square$

C Correlated competence and integrity

Replace the independent prior of the main paper with

$$\Sigma_0 = \begin{pmatrix} 1/\kappa & \sigma \\ \sigma & 1/\mu \end{pmatrix}, \qquad |\sigma| < 1/\sqrt{\kappa\mu}.$$

Let

$$d_\sigma = 1 - \kappa\mu\sigma^2, \qquad r_q = \kappa/d_\sigma, \qquad r_a = \mu/d_\sigma, \qquad r_{qa} = -\kappa\mu\sigma/d_\sigma.$$

Under an affine conjecture, write

$$y = Y - m_0 = \alpha q - ba + \varepsilon, \qquad Z = q + \nu.$$

The posterior precision matrix is

$$\Omega = \Sigma_0^{-1} + p(\alpha, -b)^\top(\alpha, -b) + \lambda(1,0)^\top(1,0) = \begin{pmatrix} L & \mathsf{n} \\ \mathsf{n} & R \end{pmatrix},$$

where

$$L = r_q + \lambda + p\alpha^2, \qquad \mathsf{n} = r_{qa} - p\alpha b, \qquad R = r_a + pb^2, \qquad \Delta_\sigma = LR - \mathsf{n}^2.$$

The manipulation first-order condition has the same linear structure after replacing the independent-prior coefficients by the correlated-prior record sensitivity s_Y and the coefficient u, but the slope-consistency equation changes. Specifically, the posterior mean of misalignment is

$$M(Y, Z) = \mathbb{E}[a \mid Y, Z] = -s_Y(Y - m_0) + s_Z Z,$$

with

$$s_Y = \frac{p\{b(r_q + \lambda) + \alpha r_{qa}\}}{\Delta_\sigma}, \qquad s_Z = \frac{\lambda(p\alpha b - r_{qa})}{\Delta_\sigma}.$$

Setting $u = s_Z - s_Y$, the expert's first-order condition gives

$$m_0 = \theta\eta s_Y, \qquad m_1 = \frac{\tau s_Y^2}{1 + \tau s_Y^2}, \qquad m_2 = \frac{\tau s_Y u}{1 + \tau s_Y^2}, \qquad \tau = \theta\gamma.$$

An affine equilibrium is characterized by the full coefficient system

$$\mathcal{F}(m_0, b, \alpha; p, \lambda, \tau, \sigma) = 0,$$

where

$$\mathcal{F} = \begin{pmatrix} m_0 - \theta\eta s_Y \\ (1 - b) - \tau b s_Y^2 \\ (\alpha - 1)s_Y - (1 - b)u \end{pmatrix}.$$

This is not the independent-prior scalar manipulation fixed point of the main paper. The correlated-prior slope equation is

$$\alpha = b + (1 - b)\frac{\lambda(p\alpha b - r_{qa})}{p\{b(r_q + \lambda) + \alpha r_{qa}\}},$$

which reduces to the independent-prior relation $\alpha = b(\kappa + \lambda)/(\kappa + \lambda b)$ only at $\sigma = 0$.

Proposition 13. *Fix $p > 0$, $\tau > 0$, and $\lambda_0 > 0$. Let $(m_{0,0}, b_0, \alpha_0)$ be an independent-prior affine equilibrium at $\sigma = 0$. Suppose*

$$\det D_{(m_0,b,\alpha)}\mathcal{F}(m_{0,0}, b_0, \alpha_0; p, \lambda_0, \tau, 0) \neq 0.$$

Then for all sufficiently small $|\sigma|$ there is a unique affine equilibrium near $(m_{0,0}, b_0, \alpha_0)$, and its coefficients are continuously differentiable in (λ, σ) locally around $(\lambda_0, 0)$.

Proof. All posterior coefficients are smooth in $(m_0, b, \alpha, \lambda, \sigma)$ on the positive-definite domain $\Delta_\sigma > 0$. The stated nonsingularity condition gives the result by the implicit function theorem. □

Proposition 14. *At a regular independent-prior affine equilibrium satisfying the nonsingularity condition of Proposition 13, let*

$$\mathcal{D}(p, \lambda, \tau, \sigma) = \frac{d}{d\lambda}\Phi(\alpha(p, \lambda, \tau, \sigma), b(p, \lambda, \tau, \sigma), \lambda, p, \sigma)$$

be the equilibrium derivative along the local affine branch. If

$$\mathcal{D}(p, \lambda_0, \tau, 0) < 0,$$

then $\mathcal{D}(p, \lambda_0, \tau, \sigma) < 0$ for all sufficiently small $|\sigma|$.

Proof. With $x = (b, \alpha)$ and $J = D_x(F_b, F_\alpha)$, implicit differentiation gives

$$\mathcal{D} = \Phi_\lambda - \Phi_x J^{-1} F_\lambda.$$

The right-hand side is continuous in σ near the regular equilibrium. A strict negative value at $\sigma = 0$ remains negative for all sufficiently small $|\sigma|$. □

For fixed $p > 0$ and fixed $\lambda > 0$, the local compression-band edges have the expansion

$$b_H(\sigma) = \tfrac{1}{2} - \underbrace{\frac{\kappa\mu\{2(2\kappa + \lambda)^2 + \lambda p\}}{\lambda p(2\kappa + \lambda)}}_{A_H} \sigma + O(\sigma^2),$$

$$b_L(\sigma) = \frac{\kappa}{\lambda + 2\kappa} + \underbrace{\frac{\kappa\mu\{8(\kappa + \lambda)^2 + \lambda p\}}{\lambda p(2\kappa + \lambda)}}_{B_L} \sigma + O(\sigma^2),$$

so the band width is $b_H(\sigma) - b_L(\sigma) = \frac{\lambda}{2(\lambda+2\kappa)} - (A_H + B_L)\sigma + O(\sigma^2)$. Positive covariance lowers the upper edge and raises the lower edge, locally narrowing the band; negative covariance widens it. These edge formulas are derived from the full coefficient system and are local in $\lambda > 0$. They are not uniform at $\lambda = 0$ and should not be used to recover the independent-prior formulas for p_N, p_0, or τ^*. Perturbed governance thresholds must be defined through their own regular boundary equations and then differentiated by the implicit function theorem; only local continuity follows without that additional boundary analysis.

D Competing record-builders

This appendix extends the baseline to $N \geq 2$ ex ante identical record-builders, where within this appendix N denotes the number of builders, and asks whether rivalry amplifies or disciplines the over-precision externality. The within-platform environment is the Gaussian model of the main paper. Builder i operates platform i and chooses record precision $p_i \geq 0$. Inside platform i the manipulation equilibrium and the regulator's platform-specific sorting-maximizing response $\lambda_i^R(p_i;\tau)$ are unchanged. Platforms interact only through user allocation. To avoid collisions with the single-builder notation, this appendix uses platform-subscripted symbols throughout, writing $R_{\mathsf{P}} > 0$ for the margin per contestable unit of platform mass, $\beta_{\mathsf{P}} > 0$ for logit demand sensitivity, and $\omega_{\mathsf{P}} \in [0,1]$ for the weight users place on integrity sorting relative to raw diagnosticity.

Platform i earns appeal

$$A_{\omega_{\mathsf{P}}}(p_i;\tau) = (1-\omega_{\mathsf{P}})\,B(p_i) + \omega_{\mathsf{P}}\,\varphi(p_i;\tau),$$

where B is the raw-record operational value of the main paper (the mutual-information or quadratic-loss value $B(p) = \frac{1}{2}\log(1+p\sigma_T^2)$) and $\varphi(p;\tau)$ is the regulator's sorting response value. Users allocate across platforms by multinomial logit demand. Platform i receives share

$$s_i(p_1,\ldots,p_N) = \frac{\exp\{\beta_{\mathsf{P}} A_{\omega_{\mathsf{P}}}(p_i;\tau)\}}{\sum_{j=1}^N \exp\{\beta_{\mathsf{P}} A_{\omega_{\mathsf{P}}}(p_j;\tau)\}}.$$

Total contestable platform mass is normalized to N, so each platform receives mass one at the symmetric allocation. Builder i's payoff is

$$\Pi_i(p_i,p_{-i}) = A_{\omega_{\mathsf{P}}}(p_i;\tau) - C_{\mathsf{D}}(p_i) + R_{\mathsf{P}} N\, s_i(p_i,p_{-i}).$$

At a symmetric profile $p_i = p$ for all i, each platform has share $1/N$ and the marginal share effect is $\partial(Ns_i)/\partial p_i = \beta_{\mathsf{P}}(N-1)/N \cdot A'_{\omega_{\mathsf{P}}}(p;\tau)$. Define the rivalry intensity

$$\varkappa_N \equiv R_{\mathsf{P}}\beta_{\mathsf{P}}\frac{N-1}{N}.$$

An interior symmetric equilibrium precision $p_N^C(\omega_{\mathsf{P}},\tau)$ satisfies the first-order condition

$$C'_{\mathsf{D}}(p_N^C) = (1+\varkappa_N)\big[(1-\omega_{\mathsf{P}})\,B'(p_N^C) + \omega_{\mathsf{P}}\,\varphi_p(p_N^C;\tau)\big].$$

An interior symmetric equilibrium under the stated selection is assumed to exist; the first-order condition is necessary and second-order or existence conditions are not verified here. Under a symmetric strict-single-crossing selection, this condition pins down the selected symmetric equilibrium, and the comparison with the monopoly precision is then a sign condition.

Proposition 15. *Suppose the separated monopoly benchmark is over-precise, $p^{\mathsf{D}} > p_N(\tau)$, where $p_N(\tau)$ is the tolerance threshold of the main paper. Define*

$$\omega^*_{\mathsf{P}}(\tau) = \frac{\varkappa_N B'(p^{\mathsf{D}})}{(1+\varkappa_N)\big[B'(p^{\mathsf{D}}) - \varphi_p(p^{\mathsf{D}};\tau)\big]}.$$

Since $B'(p^{\mathsf{D}}) > 0$ and $\varphi_p(p^{\mathsf{D}};\tau) < 0$, this threshold lies in $(0,1)$ for every $N \geq 2$, $R_{\mathsf{P}} > 0$, $\beta_{\mathsf{P}} > 0$. Under the symmetric strict-single-crossing selection, three cases obtain. If $\omega_{\mathsf{P}} < \omega^*_{\mathsf{P}}(\tau)$, rivalry amplifies and the selected symmetric equilibrium satisfies $p_N^C > p^{\mathsf{D}} > p_N(\tau)$. If $\omega_{\mathsf{P}} > \omega^*_{\mathsf{P}}(\tau)$, rivalry disciplines and $p_N^C < p^{\mathsf{D}}$. In the pure sorting-reward case $\omega_{\mathsf{P}} = 1$, the selected symmetric equilibrium does not lie in the over-precision region $p > p_N(\tau)$.

Proof. At the symmetric profile, the marginal share effect gives

$$\partial(Ns_i)/\partial p_i\big|_{\text{sym}} = \varkappa_N A'_{\omega_{\mathsf{P}}}(p;\tau).$$

Evaluating the competitive first-order condition at the monopoly precision p^{D}, where $B'(p^{\mathsf{D}}) = C'_{\mathsf{D}}(p^{\mathsf{D}})$, the sign of the competitive perturbation is

$$S_N(\omega_{\mathsf{P}},\tau) = (1+\varkappa_N)\big[(1-\omega_{\mathsf{P}})\,B'(p^{\mathsf{D}}) + \omega_{\mathsf{P}}\,\varphi_p(p^{\mathsf{D}};\tau)\big] - B'(p^{\mathsf{D}}).$$

Under single-crossing selection, $p_N^C > p^{\mathsf{D}}$ when $S_N > 0$ and $p_N^C < p^{\mathsf{D}}$ when $S_N < 0$. Setting $S_N = 0$ and solving for ω_{P} gives $\omega^*_{\mathsf{P}}(\tau)$, which lies in $(0,1)$ because $B'(p^{\mathsf{D}}) > 0$ and $\varphi_p(p^{\mathsf{D}};\tau) < 0$. The amplification and discipline claims follow from the sign of S_N relative to ω_{P} compared with $\omega^*_{\mathsf{P}}(\tau)$.

For the case $\omega_{\mathsf{P}} = 1$, on the over-precision region $p > p_N(\tau)$, $\varphi_p(p;\tau) < 0$, so the right side of the competitive first-order condition $(1+\varkappa_N)\varphi_p(p;\tau)$ is negative, while $C'_{\mathsf{D}}(p) \geq 0$. The first-order condition thus has no solution on $p > p_N(\tau)$, and the selected platform does not choose in that region. □

Competition is not intrinsically corrective. When platforms compete for users who value raw diagnostic sharpness, rivalry adds a business-stealing return to increasing $B(p)$. Each builder ignores that a sharper record raises manipulation returns after disclosure, and gains additionally from outpacing rivals. The result is a precision race, $p_N^C > p^{\mathsf{D}}$; the stakes threshold $\tau^*(p)$ falls, and separated governance enters the self-defeating band for a strictly larger set of manipulation stakes. When platforms instead win users because their published scores sort integrity well, the competitive term changes sign on the over-compressed branch. Beyond $p_N(\tau)$, additional record precision makes the platform less attractive because it worsens sorting. Rivalry then disciplines precision, and in the pure sorting-reward specification prevents over-precision altogether. The threshold $\omega^*_{\mathsf{P}}(\tau)$ marks the flip, not the mere presence of multiple builders. The characterization is for the stated logit allocation and symmetric equilibrium selection; asymmetric equilibria and alternative allocation rules are not characterized.

E Robustness of the disclosure instrument

The regulator in the main paper discloses through a single Gaussian precision signal. This appendix shows that the scalar-precision restriction is without loss within the Gaussian class and that the deconfounding force extends to general Blackwell-ordered experiments.

Proposition 16. *Suppose the regulator discloses any finite Gaussian experiment about competence, $X = hq + \xi$ with $h \in \mathbb{R}^n \setminus \{0\}$ and $\xi \sim \mathcal{N}(0,\Sigma)$, $\Sigma \in \mathbb{R}^{n\times n}$, $\Sigma \succ 0$, independent* *of (q, a, ε). The market's posterior over (q, a) depends on X only through the scalar statistic $h^\top \Sigma^{-1} X/(h^\top \Sigma^{-1} h)$ and the effective precision $\lambda_{\text{eff}} = h^\top \Sigma^{-1} h$. Scalar precision is therefore without loss within the Gaussian competence-experiment class, and the self-defeating disclosure theorem and the optimal disclosure characterization of the main paper apply with λ read as λ_{eff}.*

Proof. For $X = hq + \xi$ with $\xi \sim \mathcal{N}(0, \Sigma)$ independent of (q, a, ε), the conditional density of X given q is proportional to $\exp\{-\frac{1}{2}(X - hq)^\top \Sigma^{-1}(X - hq)\}$. The only term depending on q is

$$-\tfrac{1}{2}\big(q^2\, h^\top \Sigma^{-1} h - 2q\, h^\top \Sigma^{-1} X\big) = -\tfrac{1}{2}\lambda_{\text{eff}}\,(q - Z_{\text{eff}})^2 + \text{const},$$

with $\lambda_{\text{eff}} = h^\top \Sigma^{-1} h$ and $Z_{\text{eff}} = h^\top \Sigma^{-1} X/\lambda_{\text{eff}}$. Thus the likelihood of X about q is that of a single Gaussian signal $Z_{\text{eff}} = q + \nu_{\text{eff}}$ with $\nu_{\text{eff}} \sim \mathcal{N}(0, \lambda_{\text{eff}}^{-1})$, and since X is independent of (a, ε) given q, the posterior over (q, a) given (Y, X) equals that given (Y, Z_{eff}). The independent-vector case $Z_k = q + \nu_k$ is the special case $h = (1, \dots, 1)^\top$ and $\Sigma = \operatorname{diag}(\lambda_1^{-1}, \dots, \lambda_n^{-1})$, giving $\lambda_{\text{eff}} = \sum_k \lambda_k$. □

The same logic extends to any competence experiment, not only Gaussian ones, and locates both channels in a single statistic.

Proposition 17. *Let the regulator disclose any competence experiment $\mathcal{E}$, a signal about q whose noise is independent of (a, ε) given q and that yields a record Y with a smooth positive conditional density and equilibrium misalignment loading $b > 0$, and write $M_{\mathcal{E}} = \mathbb{E}[a \mid Y, \mathcal{E}]$ and $s_Y(\mathcal{E}) = -\mathbb{E}[\partial_Y M_{\mathcal{E}}]$ for the posterior mean of misalignment and the record sensitivity. At a fixed behavior rule under which the record loads on misalignment with net coefficient $-b$,*

$$b\, s_Y(\mathcal{E}) = \mu\ \operatorname{Var}(M_{\mathcal{E}}),$$

which reduces to the baseline $s_Y = pbK/\Delta$ with $\Phi = b\, s_Y/\mu$. If $\mathcal{E}'$ Blackwell-dominates $\mathcal{E}$, then at fixed behavior $\operatorname{Var}(M_{\mathcal{E}'}) \geq \operatorname{Var}(M_{\mathcal{E}})$.

Proof. Condition on the experiment realization $\mathcal{E} = s$ and write f_s for the conditional density of Y. At a fixed behavior rule the record's net loading on misalignment is $-b$, the structural coefficient -1 offset by the inflation slope, so conditioning on $\mathcal{E} = s$, Brown's identity gives $M_{\mathcal{E}}(y, s) = \mathbb{E}[a \mid Y = y, \mathcal{E} = s] = (b/\mu)\, \partial_y \log f_s(y)$ and $\partial_y M_{\mathcal{E}} = (b/\mu)\, \partial_{yy} \log f_s$. The score identity $\mathbb{E}[(\partial_y \log f_s)^2 \mid s] = -\mathbb{E}[\partial_{yy} \log f_s \mid s]$ and $\mathbb{E}[M_{\mathcal{E}}^2 \mid s] = (b/\mu)^2 \mathbb{E}[(\partial_y \log f_s)^2 \mid s]$ give $-\mathbb{E}[\partial_y M_{\mathcal{E}} \mid s] = (\mu/b)\, \mathbb{E}[M_{\mathcal{E}}^2 \mid s]$. Averaging over s and using $\mathbb{E}[M_{\mathcal{E}}] = \mathbb{E}[a] = 0$ yields $b\, s_Y(\mathcal{E}) = \mu \operatorname{Var}(M_{\mathcal{E}})$. For monotonicity, if $\mathcal{E}'$ Blackwell-dominates $\mathcal{E}$ then $\mathcal{E}$ is a garbling of $\mathcal{E}'$ that is independent of (q, a, Y) given $\mathcal{E}'$, so $(Y, \mathcal{E})$ is a garbling of $(Y, \mathcal{E}')$ and $M_{\mathcal{E}} = \mathbb{E}[M_{\mathcal{E}'} \mid Y, \mathcal{E}]$. Conditional expectation is nonexpansive in L^2 and both means are zero, so $\operatorname{Var}(M_{\mathcal{E}}) \leq \operatorname{Var}(M_{\mathcal{E}'})$. □

On a regular equilibrium branch the identity yields an experiment-level statement of the governance externality under three structural hypotheses. (i) Equilibrium sorting and the manipulation return depend on the experiment only through a scalar sharpness index. (ii) Sorting is single-peaked in the induced compression state. (iii) There is an upper record-diagnosticity tolerance $p_N(\tau)$ above which no experiment implements the sorting peak, with $\varphi_\tau(p) \equiv \max_{\mathcal{E}} \Phi_\tau(p, \mathcal{E})$ strictly decreasing for $p > p_N(\tau)$ and $\varphi'_{\tau,+}(p_N(\tau)) = 0$.

Proposition 18. *Consider a class of Blackwell-ordered competence experiments satisfying the three structural hypotheses above. Let* p^{D} *denote the precision chosen by a separated raw-record builder whose payoff is independent of the manipulation stakes. Under the third hypothesis, a separated raw-record builder with* $p^{\mathsf{D}} > p_N(\tau)$ *leaves the discloser's response over-compressed, meaning* $x^R_\tau(p^{\mathsf{D}}) < x^\star$*. Under the second hypothesis and right-side dominance (every over-compressed design matched by a non-over-compressed alternative with weakly higher sorting at no higher record cost), a sorting-maximizing joint controller has an optimum on the non-over-compressed side, and with strictly increasing record cost every optimum avoids over-compression. At any interior over-precise precision, the difference between the joint and separated first-order conditions is exactly*

$$\mathcal{W}_\tau(p) = -\varphi'_\tau(p) > 0,$$

the uninternalized marginal sorting value of record diagnosticity. Separated record-building sets $\rho B'(p) = C'_{\mathsf{D}}(p)$*, whereas integrated common-objective governance sets* $\rho B'(p) + \varphi'_\tau(p) = C'_{\mathsf{D}}(p)$*.*

Proof. Because $p^{\mathsf{D}} > p_N(\tau)$, the third hypothesis implies that no disclosure experiment can implement the ideal state $x^\star$; the discloser's best response $x^R_\tau(p^{\mathsf{D}})$ therefore lies on the over-compressed side. This is the first claim.

For the second claim, take any over-compressed joint design $(p, \mathcal{E})$ with $x_\tau(p, \mathcal{E}) < x^\star$. Right-side dominance supplies an alternative $(\tilde{p}, \tilde{\mathcal{E}})$ with $x_\tau(\tilde{p}, \tilde{\mathcal{E}}) \geq x^\star$, $\tilde{p} \leq p$, and $\Phi_\tau(\tilde{p}, \tilde{\mathcal{E}}) \geq \Phi_\tau(p, \mathcal{E})$. Since C_{D} is weakly increasing, the alternative has weakly higher net sorting value. With strictly increasing cost and $\tilde{p} < p$, the domination is strict. Thus a sorting-maximizing joint controller has a no-over-compression selection.

For the third claim, the separated builder's first-order condition omits $\varphi'_\tau(p)$, while the joint common-value condition includes it. Subtracting the two gives the wedge $-\varphi'_\tau(p)$. On the over-precision branch, $\varphi'_\tau(p) < 0$, so the omitted term is a positive marginal damage from additional record diagnosticity. □

The Gaussian model verifies each hypothesis in closed form (scalar reduction by Proposition 16, single-peakedness by $\Phi = \sqrt{b(1-b)/\tau}/\mu$, the tolerance by the over-precision externality theorem of the main paper, and right-side dominance by the single-designer escape proposition of the main paper). The over-precision externality is therefore not tied to Gaussian functional forms. With $\varsigma = 0$ and sorting-only pricing, the welfare wedge $\mathcal{W}(p)$ of the main paper's efficiency section reduces to the sorting wedge here.

E.1 Record composition

The scalar precision analyzed in the main paper treats the record's loading on misalignment as fixed. This subsection extends the governance externality to the fixed-composition Gaussian record family and identifies the boundary of that extension.

Proposition 19. *Consider the record family* $Y = q - \vartheta a + m + \varepsilon$ *with* $\varepsilon \sim \mathcal{N}(0, p^{-1})$ *and fixed* $\vartheta > 0$*. Under the precision bound of the main paper (condition U), the affine manipulation equilibrium on this family reduces to the baseline with relative compression state* $x = b/\vartheta$*. In particular:*

(i) Equilibrium sorting has the compact form

$$\Phi_\vartheta(x) = \frac{\vartheta}{\mu}\sqrt{\frac{x(1-x)}{\tau}},$$

with interior peak at $x = \frac{1}{2}$ and peak value $\vartheta/(2\mu\sqrt{\tau})$.

(ii) The tolerance thresholds are

$$p_0(\vartheta) = \frac{4\mu}{2\vartheta\sqrt{\tau} - \vartheta^2}, \qquad p_N(\vartheta) = \frac{4\mu}{2\vartheta\sqrt{\tau} - \vartheta^2 - \mu/\kappa},$$

with $p_0(\vartheta) < p_N(\vartheta)$ whenever the denominators are positive.

(iii) For $p > p_N(\vartheta)$ the regulator's sorting-optimal response sits at the lower edge $x^R < \frac{1}{2}$, and the response value has zero right derivative at the tolerance threshold:

$$\varphi'_{\vartheta,+}(p_N(\vartheta)) = 0.$$

The baseline results correspond to $\vartheta = 1$, where $x = b$, $p_N(1) = p_N$, and the zero derivative recovers the response-value lemma of the main paper.

Proof. Set $x = b/\vartheta$ so that $b = \vartheta x$. The posterior sensitivity becomes $s_Y = p\vartheta x K/\Delta_\vartheta$ with Δ_ϑ as in the baseline but with b replaced by ϑx. The fixed-point condition $\theta\gamma s_Y^2 = (1-x)/x$ replicates the type-dependent manipulation proposition of the main paper with this substitution. Sorting is $\Phi_\vartheta = \vartheta x s_Y/\mu$; substituting the equilibrium relation gives $\Phi_\vartheta(x) = (\vartheta/\mu)\sqrt{x(1-x)/\tau}$ with peak $\vartheta/(2\mu\sqrt{\tau})$ at $x = \frac{1}{2}$. The thresholds $p_0(\vartheta)$ and $p_N(\vartheta)$ follow the response-value lemma derivation of the main paper with $b = \vartheta x$, evaluating the fixed-point map at $x = \frac{1}{2}$. For (iii), the lower-edge branch gives $x_p^R(p_N(\vartheta)^+) = -1/(4p_N(\vartheta))$, and $\Phi_{\vartheta,x}(\frac{1}{2}) = 0$ yields $\varphi'_{\vartheta,+}(p_N(\vartheta)) = 0$. □

The baseline is $\vartheta = 1$ throughout. For any fixed $\vartheta > 0$ the governance wedge and the zero-derivative property at the tolerance extend directly, with p_N replaced by $p_N(\vartheta)$ and the analysis of the over-precision and efficiency sections of the main paper applying on each fixed-ϑ slice.

The composition-escape boundary follows from the over-precision condition $2\vartheta\sqrt{\tau} > \vartheta^2 + \mu/\kappa + 4\mu/p$. For stakes above $\mu/\kappa + 4\mu/p$ the over-precise compositions form the interval $(\vartheta_-, \vartheta_+)$ with $\vartheta_\pm = \sqrt{\tau} \pm \sqrt{\tau - (\mu/\kappa + 4\mu/p)}$. Lowering ϑ below ϑ_- escapes over-precision but reduces the peak sorting proportionally. For a raw-record builder with operational target $T = q - a$, the squared target-record correlation $\rho^2_{T,Y^0}(\vartheta)$ has derivative of sign $\kappa^{-1}(1-\vartheta) + p^{-1} > 0$ on $(0, 1]$, so private incentives favor larger ϑ, not the escape direction. The reach of the over-precision externality theorem is unaffected.

For non-collinear two-signal records, the manipulation and integrity-loading sensitivities become distinct objects; no scalar index carries the equilibrium. Collinear copies reduce to effective precision by Proposition 16; non-collinear records lie outside the scalar framework.

F Foundations for the convex penalty

This appendix derives the curvature $\gamma = -D''(\bar{M})$ from four classes of market structure. On the Gaussian-affine branch posterior variance is signal-independent, so the mean-only demand of the main paper is exact for each foundation.

Screening with heterogeneous client tolerance. If clients have a tolerance threshold with distribution F and density f and the expert earns margin π_0 per retained client, then $D(M) = d_0 + \pi_0[1 - F(M)]$ and $-D''(M) = \pi_0 f'(M)$. The foundation gives $\gamma = \pi_0 f'(\bar{M}) > 0$ exactly when the density of marginal clients is increasing at the participation margin.

Risk-averse demand with misalignment-scaled quality variance. Let delivered value be $X = \bar{x} - a + \zeta$ with $\mathrm{Var}(\zeta \mid a) = \sigma_0^2 + c_v a^2$, $c_v > 0$. For a mean-variance client with risk aversion $r_a > 0$ and expert margin $\pi_0 > 0$, the demand at signal-independent posterior variance Σ is $D(M, \Sigma) = d_0 - \pi_0 M - \frac{\pi_0 r_a c_v}{2} M^2 - \mathrm{const}(\Sigma)$, so $\gamma_{\text{risk}} = \pi_0 r_a c_v > 0$. This is the exact foundation for the quadratic reduced form.

Concave match surplus. With expert payoff $\max_r G(r, M, \Sigma)$ and $G_{rr} < 0$, the envelope theorem gives

$$-D_{MM} = H(r^*) - G_{Mr}^2/(-G_{rr}),$$

where $H(r^*) > 0$ is the direct concavity of match surplus. Curvature is positive when direct concavity dominates task-reassignment options, but sign-ambiguous in general; it is positive if the surplus is separable ($G_{Mr} = 0$).

Multiplicative trust demand. For $D(M) = \pi_0 Q_0 \exp[-\psi(M)]$ with $\psi' > 0$, one has $-D''(M) = D(M)[\psi'' - (\psi')^2]$. Under pure exponential trust $\psi(M) = rM$, this is $-D(M)r^2 < 0$. Positive local curvature requires $\psi''(\bar{M}) > (\psi'(\bar{M}))^2$, which the canonical exponential does not satisfy.

G A finite-type model

This appendix shows the self-defeating force outside the Gaussian-quadratic class. Integrity is binary, $A \in \{G, B\}$ with $\Pr(A = B) = \pi_B \in (0, 1)$; competence is $Q \in \{1, \dots, N\}$, $N \geq 2$, uniform and independent of A. A good type produces $Y = Q$; a bad type launders, producing $Y = Q$ with probability $x \in [0, 1]$ and $Y = Q - 1$ otherwise, at convex cost $\beta c(x)$ with $c'(0) = 0$ and $c'' > 0$. The reputational penalty $P(\rho_B)$ is increasing and convex in the posterior probability of the bad type. We assume $\beta c'(1) > P(1) - P(\pi_B)$, so the bad type's laundering choice is interior. The regulator discloses competence through a reveal-or-null experiment in which $Z_\delta = Q$ with probability $\delta \in [0, 1]$ and $Z_\delta = \varnothing$ otherwise, Blackwell increasing in δ.

When Q is revealed, $Y = Q - 1$ identifies the bad type while $Y = Q$ pools the good type with a laundering bad type, giving posterior $\rho_B(x) \equiv \pi_B x/(1 - \pi_B + \pi_B x)$ at the pooled record. With mutual information as the sorting measure,

$$\Phi_F(\delta, x) = D_N(\delta)\, I_R(x), \qquad D_N(\delta) = \delta + \frac{1-\delta}{N},$$

$$I_R(x) = H(\pi_B) - (1 - \pi_B + \pi_B x)\, H\big(\rho_B(x)\big),$$

where H is binary entropy, $I_R(x)$ is the revealed-competence information about integrity, strictly decreasing in laundering x, and the deconfounding factor satisfies $D'_N(\delta) = 1 - \frac{1}{N} > 0$. The product mirrors the Gaussian factorization $\Phi = b\, s_Y/\mu$, with D_N playing the role of the deconfounding sensitivity and I_R the role of the compression factor.

A single bad expert chooses x given the posterior schedule. Let $R_R(x) = P(1) - P\big(\rho_B(x)\big) > 0$ be the reputational return. The marginal return to laundering is $D_N(\delta) R_R(x)$, so the equilibrium condition is

$$\beta\, c'(x) = D_N(\delta)\, R_R(x). \tag{33}$$

The left side is increasing and the right side decreasing in x, giving a unique interior root $x^*(\delta)$ for each δ, with $dx^*/d\delta > 0$: more disclosure raises equilibrium laundering. Equilibrium sorting is $\Phi^*_F(\delta) = D_N(\delta)\, I_R(x^*(\delta))$, and its disclosure elasticity decomposes as

$$\frac{d \log \Phi^*_F}{d\delta} = \frac{D'_N(\delta)}{D_N(\delta)} - \ell_I(x^*)\, \frac{dx^*}{d\delta}, \qquad \ell_I(x) \equiv -\frac{I'_R(x)}{I_R(x)} > 0,$$

a deconfounding elasticity opposed by a compression elasticity, mirroring the balance $\Phi = b\, s_Y/\mu$ of the main model. Disclosure becomes self-defeating where the compression elasticity overtakes the deconfounding one.

Uniqueness of that crossing is governed by the hazard index

$$\Xi(x) \equiv \kappa_c(x) + \kappa_R(x) - \ell_I(x), \qquad \kappa_c(x) \equiv \frac{c''(x)}{c'(x)}, \quad \kappa_R(x) \equiv -\frac{R'_R(x)}{R_R(x)},$$

since differentiating (33) gives $dx^*/d\delta = (D'_N/D_N)/(\kappa_c + \kappa_R)$ and hence $\operatorname{sgn}(d \log \Phi^*_F/d\delta) = \operatorname{sgn} \Xi(x^*)$. Two primitive curvature conditions make Ξ cross zero once. Say the marginal cost is entropy-dominantly log-concave if $-\frac{d^2}{dx^2} \log c'(x) > [-\ell'_I(x)]_+$ on $(0,1)$, and the penalty has monotone reverse hazard if $h_P(r) \equiv (1-r)^2 P'(r)/\big(P(1) - P(r)\big)$ is weakly decreasing on $(0, \pi_B)$.

Lemma 6. *Suppose $c'(0) = 0$, $c'(x) > 0$ and $c''(x) > 0$ on $(0,1)$ with κ_c differentiable, $P(1) > P(\pi_B)$, P finite and differentiable near $[0, \pi_B]$, and h_P weakly decreasing on $(0, \pi_B)$. If the marginal cost is entropy-dominantly log-concave, then Ξ is strictly decreasing on $(0,1)$, with $\lim_{x \downarrow 0} \Xi(x) = +\infty$ and $\lim_{x \uparrow 1} \Xi(x) = -\infty$. Hence there is a unique $x_H \in (0,1)$ with $\Xi(x_H) = 0$, $\Xi(x) > 0$ for $x < x_H$, and $\Xi(x) < 0$ for $x > x_H$. If in addition h_P is differentiable with $h'_P(r) \le 0$, then $\Xi'(x) < 0$ for all $x \in (0,1)$.*

Proof. Write $\bar\pi = \pi_B$, $e(x) = 1 - \bar\pi + \bar\pi x$, and $\rho(x) = \bar\pi x / e(x)$. Differentiating $e(x) H(\rho(x)) = -\bar\pi x \log \rho(x) - (1 - \bar\pi) \log(1 - \rho(x))$ shows the two terms in ρ' cancel (using $\bar\pi x/\rho = e$ and $(1-\bar\pi)/(1-\rho) = e$), so $I'_R(x) = \bar\pi \log \rho(x) < 0$. The penalty hazard satisfies $\kappa_R(x) =$

$\frac{\bar{\pi}}{1-\bar{\pi}} h_P(\rho(x))$; since ρ is increasing and h_P weakly decreasing, κ_R is weakly decreasing, with $\kappa_R' \leq 0$ when h_P is differentiable. Let $j(x) = \kappa_c(x) - \ell_I(x)$. The entropy-dominant log-concavity condition says $-\kappa_c'(x) > [-\ell_I'(x)]_+$. If $\ell_I'(x) < 0$ this gives $\kappa_c'(x) < \ell_I'(x)$; if $\ell_I'(x) \geq 0$ it gives $\kappa_c'(x) < 0 \leq \ell_I'(x)$. In both cases $j'(x) < 0$, so j is strictly decreasing, and with κ_R weakly decreasing, $\Xi = j + \kappa_R$ is strictly decreasing. The pointwise inequality $\Xi' < 0$ follows when h_P is differentiable.

For the lower endpoint, $\rho(x) \sim \frac{\bar{\pi}}{1-\bar{\pi}} x$ as $x \downarrow 0$ gives $\ell_I(x) = O(\log(1/x))$, hence ℓ_I is integrable near zero. Because $c'(0) = 0$ and $c'(x) > 0$, one has $\int_0^\varepsilon \kappa_c(t)\,dt = +\infty$. If $\lim_{x\downarrow 0}(\kappa_c - \ell_I)$ were finite, then $\kappa_c(x) \leq \ell_I(x) + M$ near zero for some M, making κ_c integrable, a contradiction. Hence $\lim_{x\downarrow 0} j(x) = +\infty$, and since $\kappa_R \geq 0$, $\lim_{x\downarrow 0} \Xi(x) = +\infty$. For the upper endpoint, $I_R(1) = 0$ and $I_R'(1) = \bar{\pi}\log\bar{\pi} < 0$ imply $I_R(x) \sim (-\bar{\pi}\log\bar{\pi})(1-x)$, so $\ell_I(x) \sim (1-x)^{-1} \to +\infty$. The cost condition implies $\kappa_c' < 0$, so κ_c is bounded above near one; the nondegenerate penalty gap gives $\kappa_R(1^-) < \infty$. Hence $\lim_{x\uparrow 1} \Xi(x) = -\infty$. Strict decrease and the endpoint limits give the unique zero x_H. □

By Lemma 6, Ξ crosses zero at a unique $x_H \in (0,1)$, with onset

$$\delta_H = \frac{N\beta c'(x_H)/R_R(x_H) - 1}{N-1}$$

whenever the right side lies in $(0,1)$. Sorting rises for $\delta < \delta_H$ and falls thereafter, the same reversal as in the main model, now characterized in a model with binary integrity, finite competence, and a discrete record. The conditions hold on a nonempty open class, for instance any power marginal cost $c'(x) \propto x^r$ with r above a finite bound, together with any increasing convex penalty of monotone reverse hazard. Neither Gaussian information nor a quadratic penalty is needed.

G.1 A record-diagnosticity margin and a separated design game

Let the record-builder choose a diagnosticity level $r \in [\underline{r}, \bar{r}]$ with strictly increasing C^2 Blackwell index $\mathcal{Q}(r) > 0$, so the effective record sharpness is

$$\mathcal{A}(r,\delta) = \mathcal{Q}(r)\, D_N(\delta), \qquad D_N(\delta) = \delta + \frac{1-\delta}{N}. \tag{34}$$

Sorting and the bad type's laundering choice then read

$$\Phi_F(r,\delta,x) = \mathcal{A}(r,\delta)\, I_R(x), \qquad \beta c'(x) = \mathcal{A}(r,\delta)\, R_R(x), \tag{35}$$

so the hazard index Ξ governs the sign of sorting in the single scalar $\mathcal{A}$. Under the maintained curvature conditions there is a peak effective sharpness

$$\mathcal{A}_H = \frac{\beta c'(x_H)}{R_R(x_H)}, \qquad \delta_H(r) = \frac{N\mathcal{A}_H/\mathcal{Q}(r) - 1}{N-1}, \tag{36}$$

with sorting rising for $\delta < \delta_H(r)$ and falling beyond it. An integrated designer who controls (r,δ) reaches $\mathcal{A}_H$ and stays on the non-self-defeating side.

In the separated game the record-builder maximizes $B_F(r) - C_{\mathsf{D}}(r)$ with interior optimum r^{D}. Let $r_N = \mathcal{Q}^{-1}(N\mathcal{A}_H)$ be the largest diagnosticity compatible with the peak at zero disclosure. If $r^{\mathsf{D}} > r_N$, then

$$\mathcal{A}(r^{\mathsf{D}}, 0) = \frac{\mathcal{Q}(r^{\mathsf{D}})}{N} > \mathcal{A}_H, \qquad \delta_H(r^{\mathsf{D}}) < 0, \tag{37}$$

so every feasible δ sits past the onset and the regulator is pushed to $\delta = 0$. The separated equilibrium incurs sorting gap

$$L_F = \mathcal{A}_H\, I_R(x_H) - \mathcal{A}^R\, I_R(x(\mathcal{A}^R)) > 0, \qquad \mathcal{A}^R = \frac{\mathcal{Q}(r^{\mathsf{D}})}{N}. \tag{38}$$

A strictly interior regulator response requires a small transparency taste or a disclosure floor. The escape, over-precision, and welfare-gap results therefore hold under binary integrity, finite competence, and reveal-or-null disclosure, outside the Gaussian-quadratic class.

G.2 Right-side dominance in the finite-type model

In the extended model, sorting reduces to scalar effective sharpness $\mathcal{A} = \mathcal{Q}(r)D_N(\delta)$ through

$$S(\mathcal{A}) = \mathcal{A}\, I_R(x(\mathcal{A})),$$

where $x(\mathcal{A})$ solves $\beta c'(x) = \mathcal{A}R_R(x)$. Under the appendix's curvature conditions, Lemma 6 implies S has a unique peak at $\mathcal{A}_H = \beta c'(x_H)/R_R(x_H)$, with S strictly increasing for $\mathcal{A} < \mathcal{A}_H$ and strictly decreasing for $\mathcal{A} > \mathcal{A}_H$.

The following five conditions are maintained. (i) The curvature conditions of the appendix hold, so $S(\mathcal{A})$ has the unique peak $\mathcal{A}_H$. (ii) The interior laundering root is maintained on the feasible effective-sharpness range; a sufficient condition is $\beta c'(1) > \mathcal{A}_{\max}[P(1) - P(\pi_B)]$, where $\mathcal{A}_{\max} = \mathcal{Q}(\bar{r})$. (iii) The non-over-diagnostic side is feasible, meaning $\mathcal{A}_{\min} \leq \mathcal{A}_H$, where $\mathcal{A}_{\min} = \mathcal{Q}(\underline{r})/N$. The nontrivial case is $\mathcal{A}_{\min} \leq \mathcal{A}_H < \mathcal{A}_{\max}$; if $\mathcal{A}_H > \mathcal{A}_{\max}$ then no over-diagnostic design exists and dominance is vacuous. (iv) Record-building cost is weakly increasing in r, and disclosure δ is not itself part of the record-building cost. (v) $\mathcal{Q}$ is strictly increasing and continuous on $[\underline{r}, \bar{r}]$.

Every over-diagnostic design is strictly dominated by a less-diagnostic one. Take any (r, δ) with $\mathcal{A}(r, \delta) > \mathcal{A}_H$. Since $D_N(\delta) \leq 1$, $\mathcal{A}_H < \mathcal{Q}(r)$. Because $\mathcal{A}_{\min} \leq \mathcal{A}_H < \mathcal{Q}(r)$, there exists $(\hat{r}, \hat{\delta})$ with $\hat{r} \leq r$ and $\mathcal{A}(\hat{r}, \hat{\delta}) = \mathcal{A}_H$. Condition (iv) gives $C_{\mathsf{D}}(\hat{r}) \leq C_{\mathsf{D}}(r)$, and single-peakedness gives $\Phi_F(\hat{r}, \hat{\delta}, x(\mathcal{A}_H)) = S(\mathcal{A}_H) > S(\mathcal{A}(r, \delta))$, so strict right-side dominance holds. By the single-designer escape proposition of the main paper, every optimum lies on the non-over-diagnostic side.

The argument uses $\mathcal{A}_H$ as the comparator, not mirror symmetry. Three failure modes are possible. The first is $\mathcal{A}_H < \mathcal{A}_{\min}$ (no feasible right-side comparator exists); the second is non-monotone record cost (lower diagnosticity need not be cheaper); the third is violated curvature conditions (S need not be single-peaked). Without the curvature conditions, right-side dominance must be checked directly; the factorization $S(\mathcal{A}) = \mathcal{A}I_R(x(\mathcal{A}))$ alone does not deliver it.

H Extended application discussion

This appendix develops the two lead applications of the main paper's application and identification section at greater length.

Physician scorecards. A hospital system decides how granular outcome measurement is, choosing case mix, coding categories, and timing windows. A regulator then mandates a competence disclosure (risk-adjusted volume or board certification) so the market can attribute bad outcomes to conduct. The record-builder, who does not internalize the disclosure margin, over-invests in precision, and that over-precise record is what case selection and outcome timing exploit. For this setting, the curvature condition $\gamma > 0$ of the main paper is most natural when marginal clients have increasing density at the retention cutoff, consistent with the screening foundation of Appendix F. The evidence from cardiac-surgery report cards (cited in the main paper) that providers selected healthier patients after report-card introduction fits the broader logic: making a confounded record more diagnostic strengthens the incentive to game it.

Financial advice. A brokerage builds the transaction and holdings record at chosen granularity while a regulator mandates credential disclosure (a series license, fiduciary designation, or conflict-of-interest certificate). The governance split is direct. On this application, the curvature condition is most natural when perceived conflicts raise quality variance for risk-averse clients, consistent with the risk-averse demand foundation of Appendix F. It is not automatic where multiplicative trust governs demand; the multiplicative-trust case delivers negative curvature and, as the main paper notes, does not generate the self-defeating band.

Scope. Both applications require that the record-builder and the disclosure authority be separate institutions and that the record load jointly on competence and conduct (the condition $b\alpha \neq 0$ of the main paper). School value-added scores and online seller ratings share the same governance template wherever a platform sets rating precision and a regulator or aggregator publishes additional competence indicators. The identification assumption of the main paper requires only a record-technology shifter that moves precision without directly altering demand, priors, or what is manipulable.

I Additional results and omitted proofs

Notation

The following objects are used throughout this appendix. Traits are (q, a) with precisions $\kappa > 0$ and $\mu > 0$. Record precision is $p > 0$; disclosure precision is $\lambda \geq 0$; $K \equiv \kappa + \lambda$. In equilibrium the record loads on misalignment with slope $-b$ and on competence with slope $\alpha = bK/(\kappa + \lambda b)$, where $b \in (0, 1)$ solves the fixed point

$$(1-b)\,\Delta^2 = \tau\, p^2 b^3 K^2, \qquad \Delta \equiv \mu K + p\Big[\mu\big(\tfrac{bK}{\kappa+\lambda b}\big)^2 + Kb^2\Big], \tag{39}$$

with $\tau = \theta\gamma$. The record sensitivity is $s_Y = pbK/\Delta$ and the sorting index is

$$\Phi = \frac{b\, s_Y}{\mu} = \frac{1}{\mu}\sqrt{\frac{b(1-b)}{\tau}}.$$

The compact form of sorting is $\Phi = \mu^{-1}\sqrt{b(1-b)/\tau}$, peaking at $b = \frac{1}{2}$. The disclosure and record tolerance thresholds are

$$p_0(\tau) = \frac{4\mu}{2\sqrt{\tau} - 1}, \qquad p_N(\tau) = \frac{4\mu}{2\sqrt{\tau} - 1 - \mu/\kappa},$$

and $p_0 < p_N$ when both are positive, i.e., when $\tau > (1 + \mu/\kappa)^2/4$. The lower-edge function is $\ell(\lambda) = \kappa/(\lambda + 2\kappa)$. Condition (U) refers to the bound $p[\mu\alpha_{1/2}^2 + K/4] < 3\mu K$ with $\alpha_{1/2} = K/(2\kappa + \lambda)$, stated in Appendix A of the main paper. The manipulation-stakes threshold $\tau^*(p^{\mathsf{D}}) = \frac{1}{4}(1 + \mu/\kappa + 4\mu/p^{\mathsf{D}})^2$ is from Theorem 2.

I.1 The general single-designer escape

The following proposition establishes that Proposition 3 in the main paper extends beyond the Gaussian class under a right-side dominance condition on the design set.

Proposition 20. *Consider a designer who chooses a joint design d from a feasible set $\mathcal{D}$, implementing sorting $\Phi(d) = \varphi(z(d))$ with φ single-peaked at $z^\star$ and record-building cost $C(k(d))$ with C weakly increasing. Write $\mathcal{D}^- = \{d : z(d) < z^\star\}$ for the over-compressed side and $\mathcal{D}^+ = \{d : z(d) \geq z^\star\}$ for the non-over-compressed side. If right-side dominance holds (every $d \in \mathcal{D}^-$ is matched by some $\hat{d} \in \mathcal{D}^+$ with $\Phi(\hat{d}) \geq \Phi(d)$ and $k(\hat{d}) \leq k(d)$), then the integrated designer's value is unchanged by restricting attention to $\mathcal{D}^+$; an optimum can be selected on the non-over-compressed side whenever an optimum exists. If the dominance is strict in objective value, every optimum lies in $\mathcal{D}^+$.*

Proof. For any $d \in \mathcal{D}^-$, right-side dominance supplies $\hat{d} \in \mathcal{D}^+$ with $\Phi(\hat{d}) \geq \Phi(d)$ and $k(\hat{d}) \leq k(d)$. Since C is weakly increasing,

$$W(\hat{d}) = \Phi(\hat{d}) - C(k(\hat{d})) \geq \Phi(d) - C(k(d)) = W(d).$$

Hence $\sup_{\mathcal{D}} W = \sup_{\mathcal{D}^+} W$, and any optimum in $\mathcal{D}^-$ is matched by one in $\mathcal{D}^+$. Under strict dominance $W(\hat{d}) > W(d)$, so every optimum lies in $\mathcal{D}^+$. □

Gaussian symmetry supplies right-side dominance in the baseline (by the mirror argument of Proposition 3 in the main paper). The finite-type model of Appendix G supplies it under its curvature and feasibility conditions, with $\mathcal{A}_H$ as the dominating alternative, so the escape survives outside the Gaussian-quadratic class. Appendix E states the over-precision wedge for general Blackwell-ordered experiments.

I.2 The welfare-and-manipulation-cost lemma, part (i)

Lemma 2 in the main paper establishes the second part of the welfare result. The following is the omitted first part.

Lemma 7. *Let welfare be $W(\lambda) = w\,\Phi(\lambda) - G(\lambda)$, where $w > 0$ weights sorting and $G(\lambda) = \mathbb{E}[c(m^*)]$ is the expected equilibrium manipulation cost. If G is nondecreasing on $[\lambda^*, \infty)$, then the welfare-optimal disclosure level is weakly below the sorting optimum λ^*, and it is zero when w is small enough that $w\,\Phi'(\lambda) \le G'(\lambda)$ on $[0, \lambda^*]$.*

Proof. Write $W'(\lambda) = w\Phi'(\lambda) - G'(\lambda)$. For $\lambda > \lambda^*$, $\Phi'(\lambda) \le 0$ and $G'(\lambda) \ge 0$ on $[\lambda^*, \infty)$, so $W' \le 0$ and no welfare optimum lies above λ^*. If in addition $w\Phi'(\lambda) \le G'(\lambda)$ for all $\lambda \in [0, \lambda^*]$, then $W' \le 0$ throughout and $\lambda = 0$ is optimal. Near an interior sorting optimum λ^* with $\Phi'(\lambda^*) = 0$ and $\Phi''(\lambda^*) < 0$, a first-order expansion gives $\lambda_W = \lambda^* - G'(\lambda^*)/(w|\Phi''(\lambda^*)|) + o(\cdot)$, so $\lambda_W < \lambda^*$ exactly when $G'(\lambda^*) > 0$. $\square$

I.3 Finite Hermite-jet exclusion

The following proposition rules out quadratic and cubic local deviations from the unique regular affine equilibrium under a finite spectral condition on the best-response map.

The Fréchet derivative $\mathcal{L} = D\mathcal{B}[g_0]$ of the best-response map at the unique regular affine equilibrium g_0 is block-lower-triangular by total Hermite degree. The affine block is controlled by condition (U). For degree $n \ge 2$, the quotient block is

$$\mathsf{L}_n = \theta b \gamma\, S_n(P)\, D_n(c)\, D_n(d),$$

where Σ is the posterior covariance at g_0; $P = I - \Sigma\,\mathrm{diag}(\kappa, \mu)$; $c = (p\alpha\mu/\Delta,\, -s_Y)$; $d = (p\alpha\mu/\Delta,\, \mu(K + p\alpha^2)/(b\Delta))$; $D_n(v)$ is the lower-bidiagonal matrix acting on monomials $e_j^{(n)} = x_q^{n-j} x_a^j$ by $D_n(v)e_j^{(n)} = (s_Y - jv_a)e_j^{(n)} - (n-j)v_q e_{j+1}^{(n)}$; $S_n(P)$ is the symmetric-power composition of P on that basis; and $\mathcal{P}_3$ is the span of prior Hermite polynomials of total degree at most three.

Proposition 21. *Fix primitives $(p, \lambda, \kappa, \mu, \theta, \gamma)$ satisfying condition (U). Assume the equilibrium residual $F(g) = g - \mathcal{B}(g)$ is C^1 from a neighborhood of g_0 in $g_0 + \mathcal{P}_3$ into $L^2(\pi)$. Construct L_2 and L_3 from the primitives as above. If $1 \notin \sigma(\mathsf{L}_2) \cup \sigma(\mathsf{L}_3)$, then in a sufficiently small $L^2(\pi)$-neighborhood of g_0 there is no regular Bayes-confirmed equilibrium whose deviation from g_0 is a nonzero polynomial of total degree at most three. The stronger condition $\|\mathsf{L}_2\|_{G_2} < 1$ and $\|\mathsf{L}_3\|_{G_3} < 1$ implies this, where G_n is the diagonal Gram matrix of the monic degree-n Hermite basis.*

Proof. The block-triangular structure gives diagonal blocks $I - \mathsf{L}_2$ and $I - \mathsf{L}_3$ for the degree-two and degree-three quotients of $DF[g_0] = I - \mathcal{L}$. The no-unit-eigenvalue condition makes each block injective. Together with injectivity of the affine block (which follows from condition (U) and the regular root), the restriction of $DF[g_0]$ to $\mathcal{P}_3$ is injective. The assumed C^1 property then makes $F(g_0 + h) \ne 0$ for all sufficiently small nonzero $h \in \mathcal{P}_3$. The norm inequalities imply the spectral inequalities. $\square$

I.4 The two-margin benchmark

When the planner can choose both record precision and disclosure precision, the reduced surplus $S(p)$ understates the full two-margin constrained-efficient surplus whenever manipulation deadweight enters the objective.

Part 1. At an interior sorting response, $\Phi_\lambda(p, \lambda^R(p)) = 0$, giving

$$\mathcal{W}_\lambda(p, \lambda^R(p)) = -\varsigma G_\lambda(p, \lambda^R(p)).$$

Part 2. When $\varsigma = 0$, $\mathcal{W}_\lambda = 0$ for every sorting response, so $\lambda^R(p)$ satisfies the full planner's disclosure first-order condition. The reduced surplus $S(p) = U_0(p) + vV_F(\varphi(p))$ equals the full fixed-p value, and p^{C} is the full two-margin constrained-efficient precision.

Part 3. At the lower edge $\alpha = \frac{1}{2}$ and $b_\lambda = 0$, so the behavioral term in G_λ vanishes and only the competence-slope component remains:

$$G_\lambda(p, \lambda^R(p)) = -\frac{\kappa b^R(1-b^R)^2}{\theta(\kappa + \lambda^R b^R)^3} < 0.$$

Hence $\mathcal{W}_\lambda = -\varsigma G_\lambda > 0$ for $\varsigma > 0$. □

I.5 Stationarity and primitive curvature

Proposition 22. *Fix a regular local-tangent response problem at an upper tolerance threshold p_N. Let $r^*(p)$ be the reference reputation on the local peak branch, let*

$$\Gamma(p) = \gamma_R(r^*(p)), \qquad \tau_R(p) = \theta\Gamma(p),$$

and suppose the local-tangent sorting value is $\Psi_{LT}(b, p) = \mu^{-1}\sqrt{b(1-b)/\tau_R(p)}$ with interior peak $b^* = 1/2$.

(i) The peak sorting value is

$$V(p) = \frac{1}{2\mu\sqrt{\tau_R(p)}},$$

and its right derivative at p_N satisfies

$$V'_+(p_N) = -\frac{1}{2}V(p_N)\frac{\Gamma'_+(p_N)}{\Gamma(p_N)}.$$

Peak-value stationarity holds if and only if $\Gamma'_+(p_N) = 0$. When $(r^*)'_+(p_N) \neq 0$, this is equivalent to $\gamma'_R(r_N) = 0$, equivalently $\mathbb{E}[-D'''(r_N + \xi)] = 0$. Under peak-value stationarity, the sorting-damage-only marginal threshold charge is zero by Theorem 3.

(ii) Suppose the smoothing noise ξ is symmetric around zero and $\gamma_R(r) = \mathbb{E}[-D''(r + \xi)]$. If $D \in C^3$, $D' < 0$, $-D'' > 0$, and the primitive curvature $C_D(M) = -D''(M)$ is nonconstant and even around $r_N = r^*(p_N)$, then D is non-quadratic and satisfies peak-value stationarity at p_N. Hence the zero sorting-damage-only marginal threshold charge extends to this non-quadratic class.

Proof of Part (i). Since the peak is at $b = 1/2$ for every p, $V(p) = (2\mu\sqrt{\tau_R(p)})^{-1}$ and differentiating gives the displayed formula. The equivalence $V'_+(p_N) = 0 \iff \Gamma'_+(p_N) = 0$ is immediate. When $(r^*)'_+(p_N) \neq 0$, the chain rule reduces stationarity to $\gamma'_R(r_N) = \mathbb{E}[-D'''(r_N + \xi)] = 0$.

Proof of Part (ii). C_D even implies C'_D odd, so $\gamma'_R(r_N) = \mathbb{E}[C'_D(r_N + \xi)] = 0$ by odd-function cancellation with symmetric ξ. Part (i) then gives $V'_+(p_N) = 0$. Nonconstancy of C_D rules out quadratic D. (Concrete member: $D_A(M) = d_0 - A\log(1 + e^{M-r_N})$ has curvature $Ae^{M-r_N}/(1 + e^{M-r_N})^2$, even around r_N.) □

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